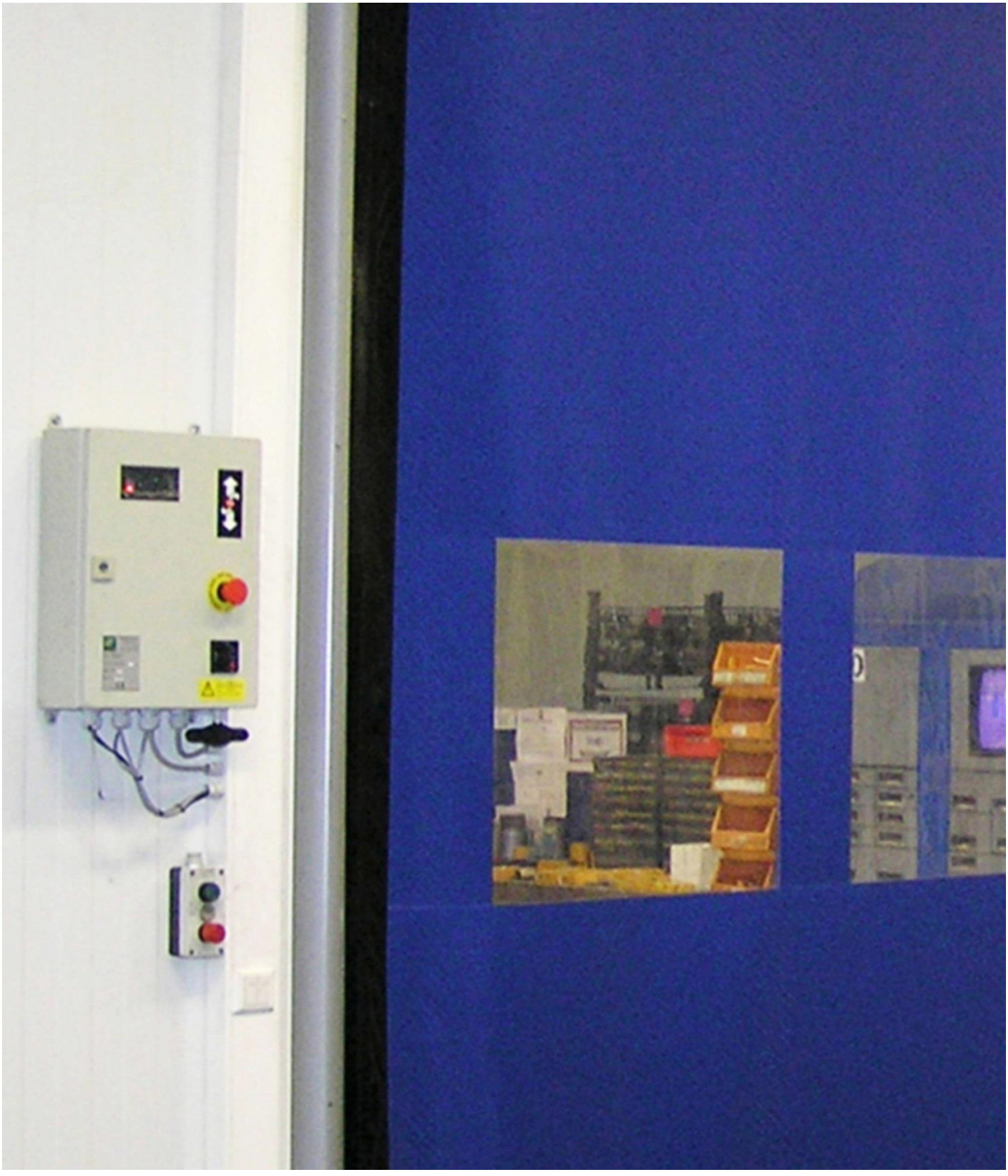
Installation manual

High speed door
ASSA ABLOY 820 DCS

ASSA ABLOY

ASSA ABLOY Entrance Systems

The global leader in
door opening solutions



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2. Introduction

Since 1987, ASSA ABLOY has concentrated its know-how on a unique and patented system of automatic, high-speed doors that have the following characteristics:

- Perfect seal
- Total safety
- Guaranteed to be self-repairing
- Wind proof
- Very high speed
- Smooth operation

ASSA ABLOY continuously develops its products to accurately meet the specific needs of every sector of the industry.
ASSA ABLOY has an experienced team for design, customer advice, production, and providing service with the shortest possible lead times.

In the context of this policy of continuous development of its products, ASSA ABLOY reserves the right to change the characteristics of its products or parts, without prior notice.

3. Directives and standards

The following guidelines and standards are implemented in the construction of ASSA ABLOY doors.

2006/42/EC	Machine Directive.
(EU) No 305/2011	Construction product regulation.
2004/108/EC	Electro Magnetic Compatibility Directive.
EN 13241-1	Industrial, commercial and garage doors without fire and smoke resistant characteristics.

ASSA ABLOY doors can be controlled in various ways. The control units offer a number of options for connecting hand operated and/or automatic opening controls.
The choice of opening commands depends on a large number of factors, including intensity of use, safety of users, the environment and the type of traffic using the door.

ASSA ABLOY Europe offers a wide range of opening controls. The installer, in consultation with the user, can offer added value within his or her area of responsibility by selecting the most appropriate actuators.

The opening controls selected by the installer or end user must meet the guideline for Electro-magnetic compatibility, **2004/108/EC.** For remote control using a radio frequency, guideline R&TT E, **99/5/EC**, also applies. The opening controls must be adapted for the use and environment in which they are to be installed. They may not influence the control unit. Neither must their operation in turn be impacted by environmental factors nor by the control unit. The control and safety components that are built-in as standard, must be replaced with identical replacement parts.

All components connected to the door, the connections, the working methods and materials must meet the required standards listed under the guidelines set out above, or those imposed by local authorities.

4. Safety instructions

4.1. General instructions

Please carefully read the safety instructions and the manual before starting any work.

The description below uses symbols to draw the reader's attention to the various dangers and to provide useful advice.

	Indicates a potential danger to persons; please take all possible precautions against risks associated with working with electrical materials, it may have power connected.
	Please follow exactly; ignoring this advice can cause a fault or a dangerous situation.
	Important information.
ASSA ABLOY	Only to be performed by ASSA ABLOY-certified persons.
	May only be performed by a technician who is authorised to operate a forklift.

4.2. Precautions during installation

- Electrocution may lead to death. Never touch parts that are 'live'. Be very careful during the installation and maintenance of electrical components of the door.
- For the safety of the user and the proper operation of the door, the door must be installed in accordance with the instructions contained in this manual.

- Only authorised personnel may work on ASSA ABLOY doors. Personnel concerned must be properly informed about the contents of the manual, remarks and warnings regarding transport, and about the installation and commissioning of this door. Work by unauthorised personnel can bring about risks for the users. Install the door with the assistance of qualified forklift drivers and electricians.
- Installation by unauthorised personnel may lead to damage of the door components and poor operation of the door.
- Not following the manual during installation, or installation by unauthorised personnel will lead to an invalidation of the warranty.
- Please ensure that all accessories meet IEC standards.
- The equipment and accessories must be installed in accordance with the manufacturer's instructions, as well as national and other standards.

4.3. Precautions during operation

- Electrocution may lead to death. Never touch parts that are 'live'. Be very careful during the installation and maintenance of electrical components of the door.
- For the safety of the user and the proper operation of the door, the door must be used in accordance with the instructions contained in this manual.
- Only use the door when the control box is locked.
- Ensure the safety warnings, covers and protections are kept in good condition. Safety warnings must remain visible at all times.
- Avoid all contact with moving parts.
- Sometimes, safety equipment needs to be removed for maintenance. Reinstall them after commissioning.

4.4. Precautions during maintenance

- Electrocution may lead to death. Never touch parts that are 'live'. Be very careful during the installation and maintenance of electrical components of the door.
- Only personnel certified by ASSA ABLOY are allowed to carry out maintenance on ASSA ABLOY doors. If maintenance is carried out by persons who have not been trained by ASSA ABLOY, the warranty will be invalidated.
- All parts used in ASSA ABLOY doors have been designed for this application. Only use original ASSA ABLOY parts.
- Sometimes, safety equipment needs to be removed for maintenance. Reinstall them after commissioning.
- In case of intervention (with regard to the electrical and/or mechanical section of the installation), the power supply must be interrupted and locked.

	Beware: the cables between main switch and power supply always remains live.

4.5. Precautions for the frequency inverter



The frequency inverter.

- When a frequency inverter has power supplied to it, the electrical elements and a number of operating controls are also 'live'. Never touch these elements. This is extremely dangerous.
- When the emergency stop is activated, the frequency inverter remains 'live'. If this is a threat to the safety of the staff, the power circuit must be interrupted by locking the main switch on the control box.
- After locking the main switch, it is always necessary to wait **15 minutes** before starting work. This is the time required for discharging the capacitors of the frequency inverter.
- The frequency inverter has integrated safety systems for stopping the door. A mechanical blockage, fluctuations in voltage and interruptions of the power supply can also bring the door to a stop.
  This is shown with a fault message on the frequency inverter screen.
  Lock the main switch before removing the cause of a blockage.
  Unlock the main switch to put the door into service again.
  An impulse via the ‘'Open' push button on the control box will reset the controller.
- For more detailed information, please check the manual of the frequency inverter in the control box.

4.6. Automatic opening during power failure

	For doors with the 'automatic opening during power failure' option, it is not enough to interrupt the general feed from the electricity supply, because the UPS current will continue to flow. In order to carry out work on the door, the door must be shut down completely by locking the main switch on the control box. The cable that connects the UPS with the main switch will then remain 'live'.

Consult the manual of the UPS for interrupting the current on the cable between the UPS and the main switch.

4.7. AUTO/MANUAL switch (dead man's operation) (Optional)

The door will close based on the dead man principle in the manual (Manu) mode: the Close pushbutton must remain pressed. The photocell and the WDD remain active. A single press of the 'open' button will open the door again.

4.8. Safety instructions

You must observe the important safety instructions given below such as the instructions for assembly and connecting when commissioning and using the control.

- All work related to the installation, commissioning and maintenance may only be performed by qualified (BA4 – BA5) personnel. The following instructions, in particular, must be observed: EN 50110, EN 60204, EN 50178, EN 60335, instructions related to the prevention of fire and accident prevention as well as the related instructions for industrial doors (EN12453, EN12978).
- The control may only be opened when all poles of the provision have been disconnected.

- If the contacts with NO contact of the relay outputs of the other terminals of the external power supply are supplied with power (for example, when a dangerous power supply is used that may still be connected once the control is disconnected or when it is unplugged), a corresponding warning label must be clearly visible on the housing of the control. ("ATTENTION! All power supplies be disconnected before accessing the terminals.").
- You must not use the control if it has been opened.
- You can only use the control with a disassembled CE plug if the power supply is controlled by a corresponding switch where all poles of the control can be separated. The plug of the replacement switch must be easily accessible.
- If the connecting cable of the device is damaged, the manufacturer, the relevant customer services or a qualified engineer must replace it to ensure dangerous situations are avoided.
- Dangerous voltage will continue to be present on the interstage circuit capacitor for another five minutes after the power supply has been disconnected. The time required for discharging until you obtain a voltage value below 60 V DC is 5 minutes at most. It is dangerous to touch the internal control components during the discharge time.
- The discharge time of the interstage circuit capacitors may be considerably lengthened to attain a voltage value below 60 VDC when the switched-mode power supply is faulty. This may mean that the time required for the discharge may be 10 minutes at most.
- The switched-mode power supply will not switch on although the interstage circuit capacitors will have been charged when there is a short-circuit or when the 24 V control voltage is extremely overloaded. Display and LEDs remain dark. The power supply will only switch on after the short-circuit or extreme overloading situation has been solved.
- The supply adapter will continue to be supplied from the interstage circuit capacitors for a few seconds after the power supply has been deactivated, which means that the power supply will continue to be active for a period of time depending on the load of the power supply.
- The processor circuit with 7-segment display, EPROM and multiplexor is in direct galvanic contact with the power supply. This must be taken into account should any control measurements be performed. (Do not use measuring equipment with a PE reference of the measuring circuit when measuring the processor circuit elements.)
- You must never use the control without appropriate earthing. Dangerous high-voltages form when the earthing is not connected on the control housing due to discharging capacity. The connection of the earthing must be implemented in accordance with EN50178 article 5.2.11.1 for safe impedances >3.5 mA.
- A bedewed (i.e. affected by moisture) control may not be activated or used. This may lead to corrosion of the control.
- If controls are used outside the specified temperature range, a check must take place through a regulated and monitored heating system to verify whether the specified operating temperature is being maintained when the provision is switched on as well as when the control is being used.
- The control must never be used with a damaged membrane button section or display window. Damaged button sections and windows must be replaced. The button block must not come in contact with sharp objects to ensure it cannot be damaged. The button section may only be manually operated.
- You must check whether the analysis cards (plug-in module) have been plugged into the correct place the first time you activate the power supply. The control may be damaged when the card has been incorrectly placed or has been twisted. The same applies when installing products of third parties that have not been approved for release.
- You must check whether the door range can be viewed by the user when there is door movement during dead man's use because the safety settings such as the tracer and photocell do not work for this type of use.
- The setting of the parameters and the function of the safety settings must be checked. Parameters, bridges and other operating elements may only be set by qualified personnel.



Not following safety measures may endanger health or may damage the control.

These safety measures must not be deemed to be exhaustive. Please contact your supplier should you have any questions about the product.

The manufacturer has carefully checked the hardware and software of the device as well as the product documentation.

The manufacturer cannot, however, fully guarantee that they will be completely free from errors.

A type designation of the device (type label that provides the name and address of the manufacturer, serial number, type designation, power supply and temperature range) must be followed by the user.

5. Technical characteristics of controllers

5.1. Single phase 230VAC

Housing dimensions (W x H x D)	300 x 400 x 120 mm without wall bracket. 500 x 395 x 210 mm without wall bracket for stainless steel.	
Fixation:	Vertical using wall brackets on housing.	
Supply voltage through L, N, PE:	230 V_{AC} ±10%, 50...60 Hz Permissible range: 180...240V ± 10% / 50...60Hz. Protection: 16A K-type.	
Internal Power consumption:	Max. 40 W with full equipment and motor not running.	
External supply Voltage1 (230 V):	230 V_{AC} ±10%, 50...60 Hz (fused on the circuit board: F202/3.15 AT)	
Control voltage/external supply 2:	24 V_{DC} regulated (±5% at nominal voltage 230 V) Max. 500 mA incl. optional plug-in module With circuit breaker, short circuit protected by central switching regulator.	
Control voltage/external supply 3:	For encoder and bottom edge Nominal voltage 11.5 V/max. 130mA	
Control inputs:	24 V DC/type 15 mA, max. 26 V DC/20 mA All inputs are to be connected potential free or: < 5 V: inactive → logical 0 > 7 V: activate → logical 1 Min. signal duration for input control commands: > 100 ms. Galvanically isolated via optocouplers on the circuit board.	
Inputs INK 1 and INK 2:	For two 24 V active 90° offset pulse inputs max. 20mA load. < 5V: inactive → logical 0, > 16 V activate → logical 1. Limit frequency: 1kHz.	
RS485 A and B:	Only for encoder. RS485 level, closed with 100 Ω.	
Safety chain / Emergency-Stop	All inputs are to be connected potential free. Contact load capacity: ≤ 26 VDC / ≤ 120 mA. When the safety chain is interrupted, no movement of the drive is possible not even in dead man mode.	
Bottom edge input:	For electrical bottom edge with 1.2 kΩ or 8.2 kΩ termination resistor and for dynamic optical systems.	
Relay outputs	If inductive loads are switched (i.e.: additional relays), these must be equipped with recovery diodes and appropriate noise suppression means (regenerative diodes varistors, RC modules).	
Relay K3: **Standard brake relay:**	Changeover contact for enabling electromechanical brakes with interposed brake rectifier. 230 V AC / 3 A. When E-stop tripped, the brake output immediately becomes active.	
Relays K4, K5, K6, K7, K8 and K9 "Fault / Door position messages / Lamp functions"	Changeover contact potential free min. 10 mA max. 230 V AC / 3 A	230V permitted only when connecting the same phase as the supply voltage. Contacts used once for power switching can no longer switch low current

Drive output:	For drives up to 0.75 kW (version -A) 1.5 kW (version -C) at 230 V. Motor constant current at 100% duty factor and 40°C ambient temperature: 5A. Motor constant current at 60% duty factor and 50°C ambient temperature: 5A. version –A: 5A, version –C: 10A Short time overload capacity: up of 15A and 22A respectively for up to 0.5 s. Max. length of the motor cable: 30 m.
Brake resistance load (optional):	max. 1.5 kW for max. 0.5 seconds. Repetition rate min. every 20 seconds.
Temperature range operating: **Storage: :**	-10...+50°C -25...+70°C
Relative humidity	Up to. 80% non-condensing
Vibration	Low vibration installation, i.e: on a concrete wall
Enclosure rating	IP54 (only in combination with the housing)
Weight	Approximately 6.5 kg

Directives		**Standards:**	
EMC Directive:	2004/108/EC	EN 50081-1/03.93: EN 50081-2/03.94: EN 61000-6-2/2001 : :	Noise emission, residential. Noise emission, industrial. Noise immunity, Industrial
Low-voltage directive:	2006/95/EC	EN 60335-1/2003:	Safety of household and similar appliances/part 1
Type-tested according to:		EN12453/2001:	Industrial, commercial and garage doors and gates safety in use of power operated doors and gates – Requirements
		EN12445/2001:	Industrial, commercial and garage doors and gates – Test procedures
Applied national technical specifications related to the above directives		EN12978/2003:	Doors and gates - Protection devices – Requirements and test procedures

5.2. Three phase 400VAC

Housing dimensions (W x H x D)	300 x 400 x 120 mm without wall bracket.	
Fixation:	By means of wall attachment at the top and bottom of the housing	
Power supply via L1, L2, L3, N, PE:	400 V_{AC} ±10%, 50...60 Hz Permissible range: 200...400V ± 10% / 50...60Hz. Protection: 16A K-type.	
Internal Power consumption:	Max. 50 W with full equipment and motor not running.	
Control voltage/external supply 2	24 V_{DC} regulated (±5% at nominal voltage 230 V) Max. 500 mA incl. optional plug-in module With circuit breaker, short circuit protected by central switching regulator.	
Control voltage/external supply 3	For encoder and bottom edge Nominal voltage 11.5 V/max. 130mA	
Control inputs:	24 V DC/type 15 mA, max. 26 V DC/20 mA All inputs are to be connected potential free or: < 5 V: inactive → logical 0 > 7 V: activate → logical 1 Min. signal duration for input control commands: > 100 ms. Galvanically isolated via optocouplers on the circuit board.	
Inputs INK 1 and INK 2:	For two 24 V active 90° offset pulse inputs max. 20mA load. < 5V: inactive → logical 0, > 16 V activate → logical 1. Limit frequency: 1kHz.	
RS485 A and B:	Only for encoder. RS485 level, closed with 100 Ω.	
Safety Chain / Emergency Stop	All inputs are to be connected potential free. Contact load capacity: ≤ 26 VDC / ≤ 120 mA. When the safety chain is interrupted, no movement of the drive is possible not even in dead man mode.	
Bottom edge input:	For electrical bottom edge with 1.2 kΩ or 8.2 kΩ termination resistor and for dynamic optical systems.	
Relay outputs	If inductive loads are switched (i.e.: additional relays), these must be equipped with recovery diodes and appropriate noise suppression means (regenerative diodes varistors, RC modules).	
Relay K3: **Standard brake relay:**	Changeover contact for enabling electromechanical brakes with interposed brake rectifier. 230 V AC / 3 A. When E-stop tripped, the brake output immediately becomes active.	
Relays K4, K5, K6, K7, K8 and K9 "Fault / Door position messages / Lamp functions"	Changeover contact potential free min. 10 mA max. 230 V AC / 3 A	230V permitted only when connecting the same phase as the supply voltage. Contacts used once for power switching can no longer switch low current

Drive output:	For drives up to 2, 2 kW (version -A) 4 kW (version -C) at 400 V. Motor constant current at 100% duty factor and 40°C ambient temperature: 5A. Motor constant current at 60% duty factor and 50°C ambient temperature: 5A. version –A: 5A, version –C: 10A Short time overload capacity: up of 15A and 22A respectively for up to 0.5 s. Max. length of the motor cable: 30 m.
Brake resistance load (optional):	max. 1.5 kW during a max. of 0.5 seconds. Repetition ratio at least every 20 seconds.
Temperature range operating: **Stored:**	-10...+50°C -25...+70°C
Relative humidity	Up to 80% non-condensing
Vibration	Low vibration-installation, i.e.: on a concrete wall
Enclosure rating	IP54 (only in combination with the housing)
Weight	Approximately 6.5 kg

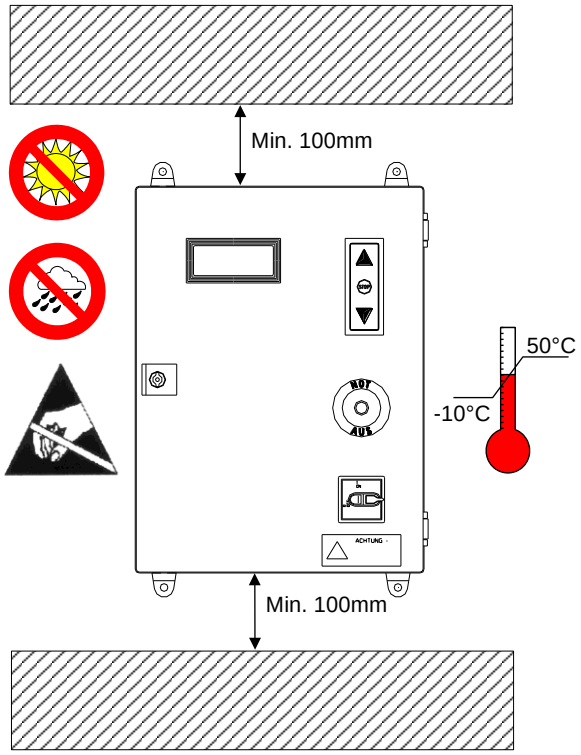
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Low-voltage directive:	2006/95/EC	EN 60335-1/2003:	Safety of household and similar appliances/part 1
Type-tested according to:		EN12453/2001:	Industrial, commercial and garage doors and gates safety in use of power operated doors and gates – Requirements
		EN12445/2001:	Industrial, commercial and garage doors and gates – Test procedures
Applied national technical specifications related to the above directives		EN12978/2003:	Doors and gates - Protection devices – Requirements and test procedures

6. Installation

6.1. Installing the control box

The device must have been switched off when the control is being installed (no current)

- The control must be checked to ensure there are no transportation or other damage before installation. Damage to the internal part of the control may under certain circumstances lead to considerable damage to the control or may represent a risk to the user's health.
- Never touch the electronic components, in particular, the processor circuit. The electronic components may be damaged or disrupted due to electrostatic discharges.
- Ensure the metal shavings that are on the control cannot fall into the housing when opening the cover of the housing.
- Ensure that the control is installed non-machined.
- Cable guides that have not been used must be closed using the correct measures to guarantee the IP54 safety type of the housing.
- The cable guides may not be exposed to a load or strain.



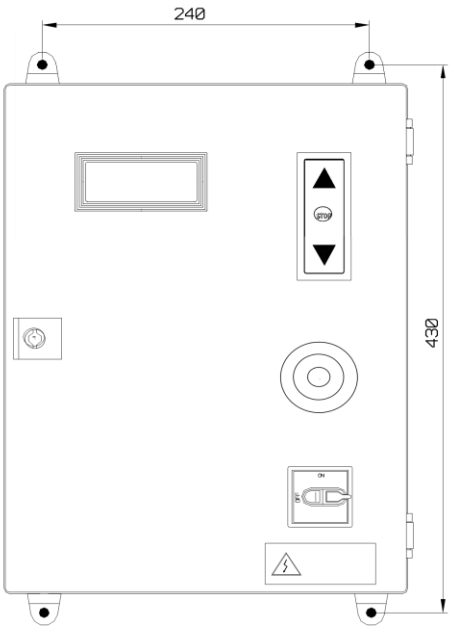


Figure 6-1: Installing the control box.

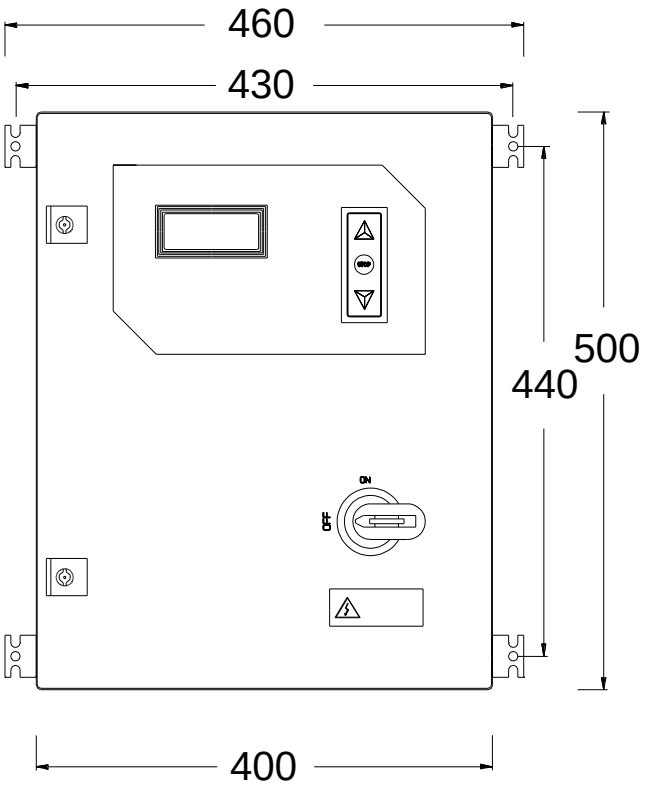


Figure 6-2: Installing the Stainless Steel control box.

6.2. Electrical connection



- When the control box is open, connection, checking and maintenance work may only be performed if the mains power has been disconnected. Please pay particular attention to the points shown in the section on “Safety Rules”.
- After disconnecting the control box, it will still carry a dangerous voltage for up to 5 minutes.
- Touching the electronic components is dangerous due to the presence of residual voltage.
- The control box must never be used when the cover is open.



- If you want to activate the control box for the first time after completing the cabling work, you must first check whether all motor connections (control side and motor side) have been correctly attached and whether the motor has been star or delta switched. Loose motor connections will, in principle, lead to damage of the frequency converter.
- All inputs to the control circuit are galvanically separated from the power supply through reinforced insulation. We recommend using an additional insulation with a test voltage of > 230 V for all components to be connected to the control box.
- For compliance with the EMC Directive, only shielded, separate motor cables may be used, whereby the shield must be connected at both ends (motor side and the control box side), and no other voltages may be carried by the cable. Maximum cable length: 30 m.
- Quick-roll doors, in particular, will generate very high electrostatic charges. The control box may be damaged as a result of the discharge of this voltage. This is why the correct precautionary measures must be taken, so that electrostatic discharge may be avoided.
  We recommend using a correctly folding ferrite via the motor cable directly to the control box, to ensure protection of the controls.
- Please check the cable diameters to be used in the data sheets.
- Max. connection diameters of the printed circuit board terminals:

	Single wire (rigid)	Fine wire (with/without core sleeve)
Screw terminals	4	2.5
Plug-in terminals	2.5	2.5
Motor terminals	6	4
Power supply	6	4

6.2.1. Connection of the power supply

6.2.1.1. Single phase 230VAC

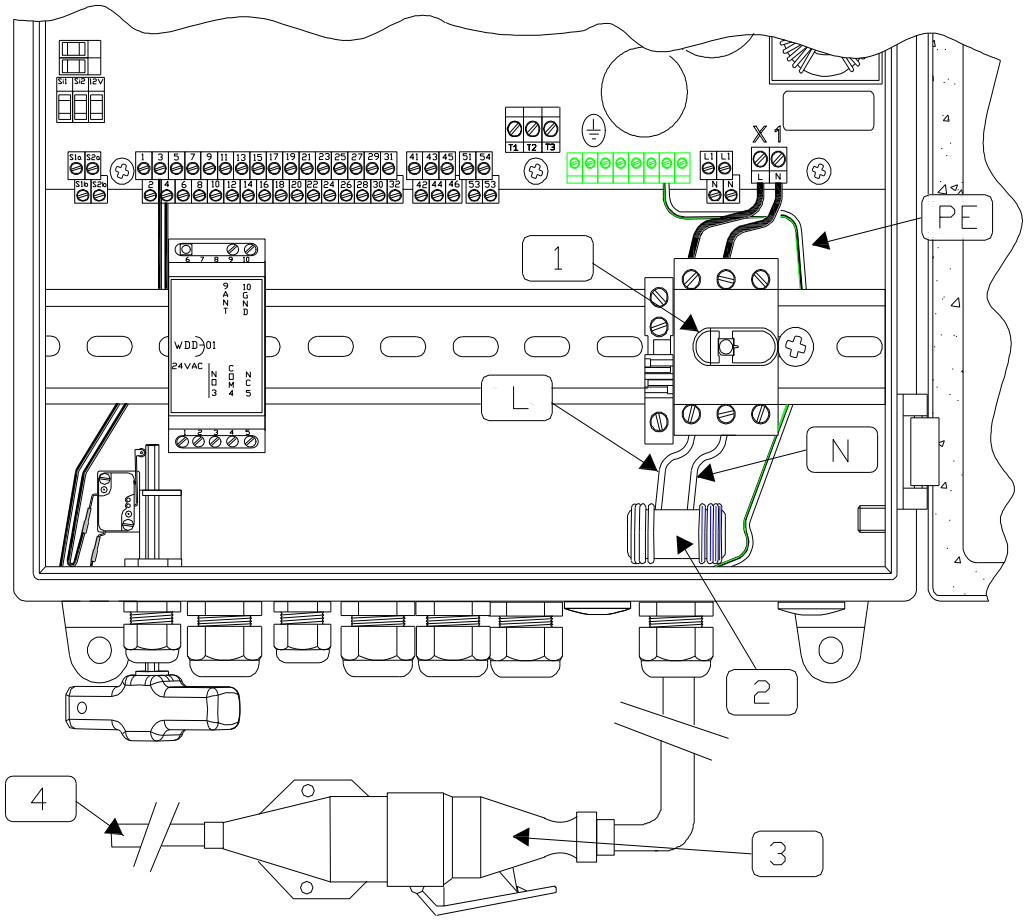


Figure 6-3: Connection of the power supply Single phase 230VAC

	The optional CE plug must remain visible and accessible from the control box location.

1	Main switch.	3	CE plug 2 poles + PE 230 V blue. OPTIONAL
2	Wind each separated L and N cables 5x through the ferrite core.	4	K-type fuse (To be installed by the client in accordance with the technical description of the door.)
	Slide the ferrite core up to the cable guide.		
	Attach the L and N cables to the main switch.		
	Lay the PE cable past the core and clamp.		

For compliance with the EMC standard, conductors L & N must be wound 5 times around the ferrite core. The windings must remain separated. The earth conductor must be connected directly to the PE terminal.

6.2.1.2. Three phase 400VAC

6.2.1.2.1. Without Neutral

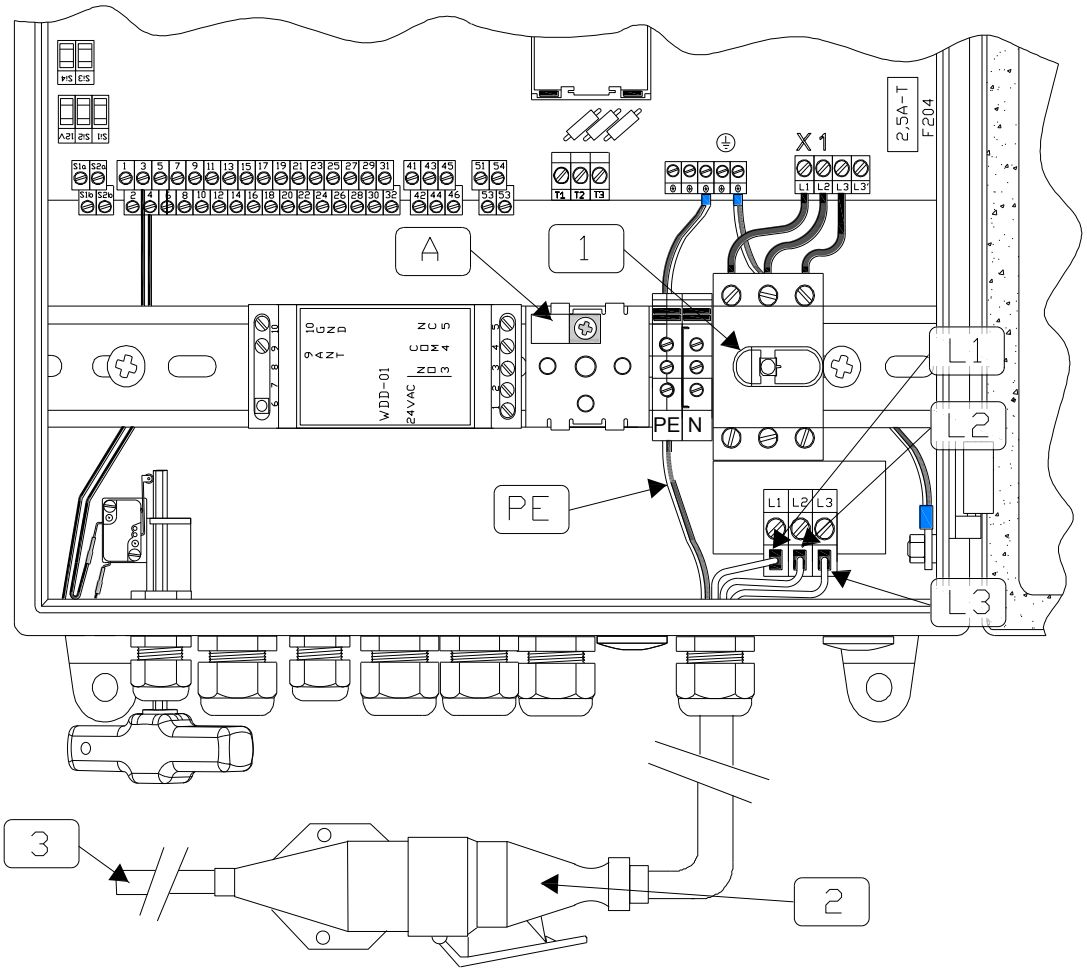


Figure 6-4: Connection of the power supply Three phase 400VAC without neutral

	The optional CE plug must remain visible and accessible from the control box location.

1	Main switch.	3	K-type fuse (To be installed by the client in accordance with the technical description of the door.)
2	CE plug 4 poles + PE red. OPTIONAL Attach the L1, L2 and L3 cables to the Filter .	A	Motor cable shield clamp

For compliance with the EMC standard, don't remove the filter connected to the input of the main switch

6.2.1.2.2. With Neutral

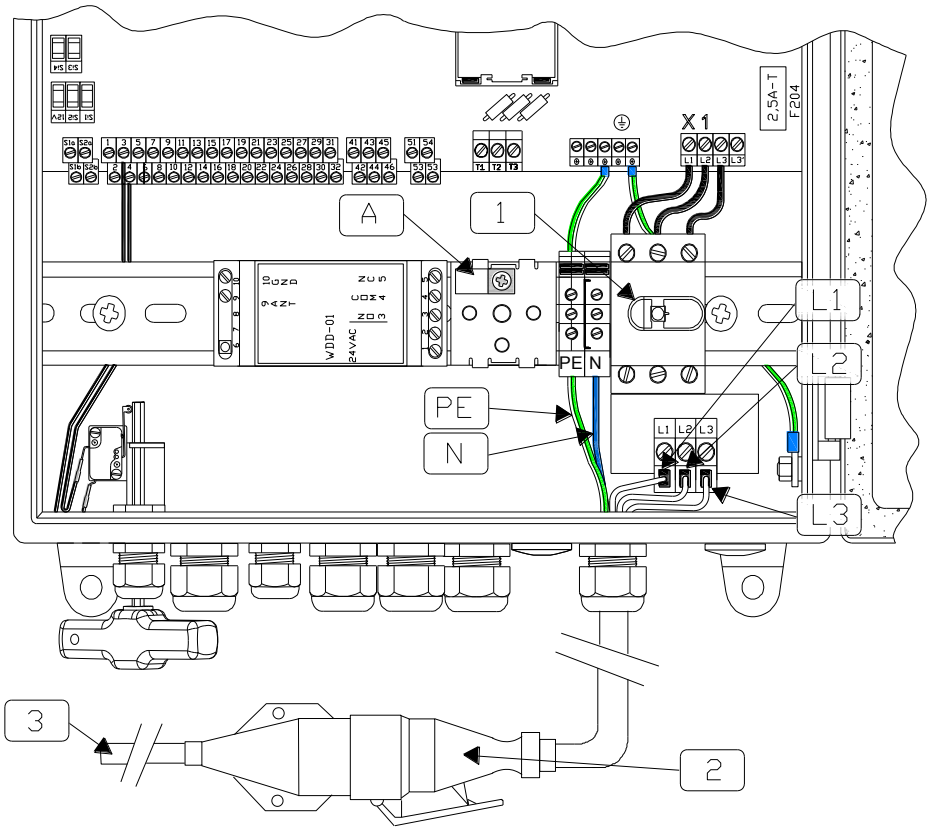


Figure 6-5: Connection of the power supply Three phase 400VAC with neutral`

	The optional CE plug must remain visible and accessible from the control box location.

1	Main switch.	3	K-type fuse (To be installed by the client in accordance with the technical description of the door.)
2	CE plug 4 poles + PE red. OPTIONAL	A	Motor cable shield clamp
	Attach the L1, L2 and L3 cables to the Filter .	N	Attach the blue cable (N) to terminal N

For compliance with the EMC standard, don't remove the filter connected to the input of the main switch

6.2.2. Motor connection for door HS9030G and HS9020G

6.2.2.1. Single phase 230VAC

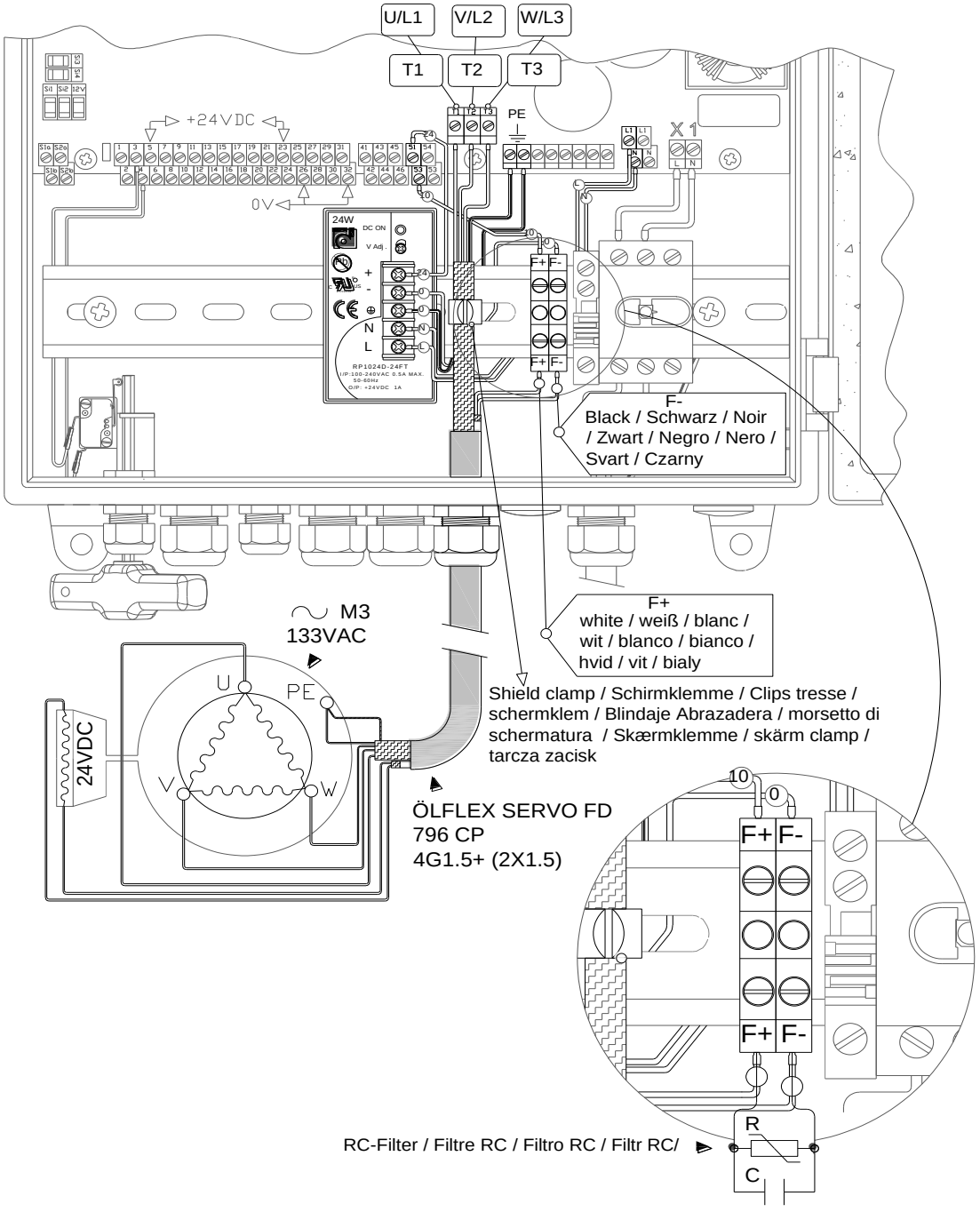


Figure 6-6: Motor connection for door HS9030G and HS9020G single phase 230VAC.

	To guarantee the trouble-free operation of the 820 DCS controller, a shielded cable must be used. Furthermore, no other wiring may be installed except the wiring for the connection of the motor.

6.2.2.2. Three phase 400VAC + Neutral

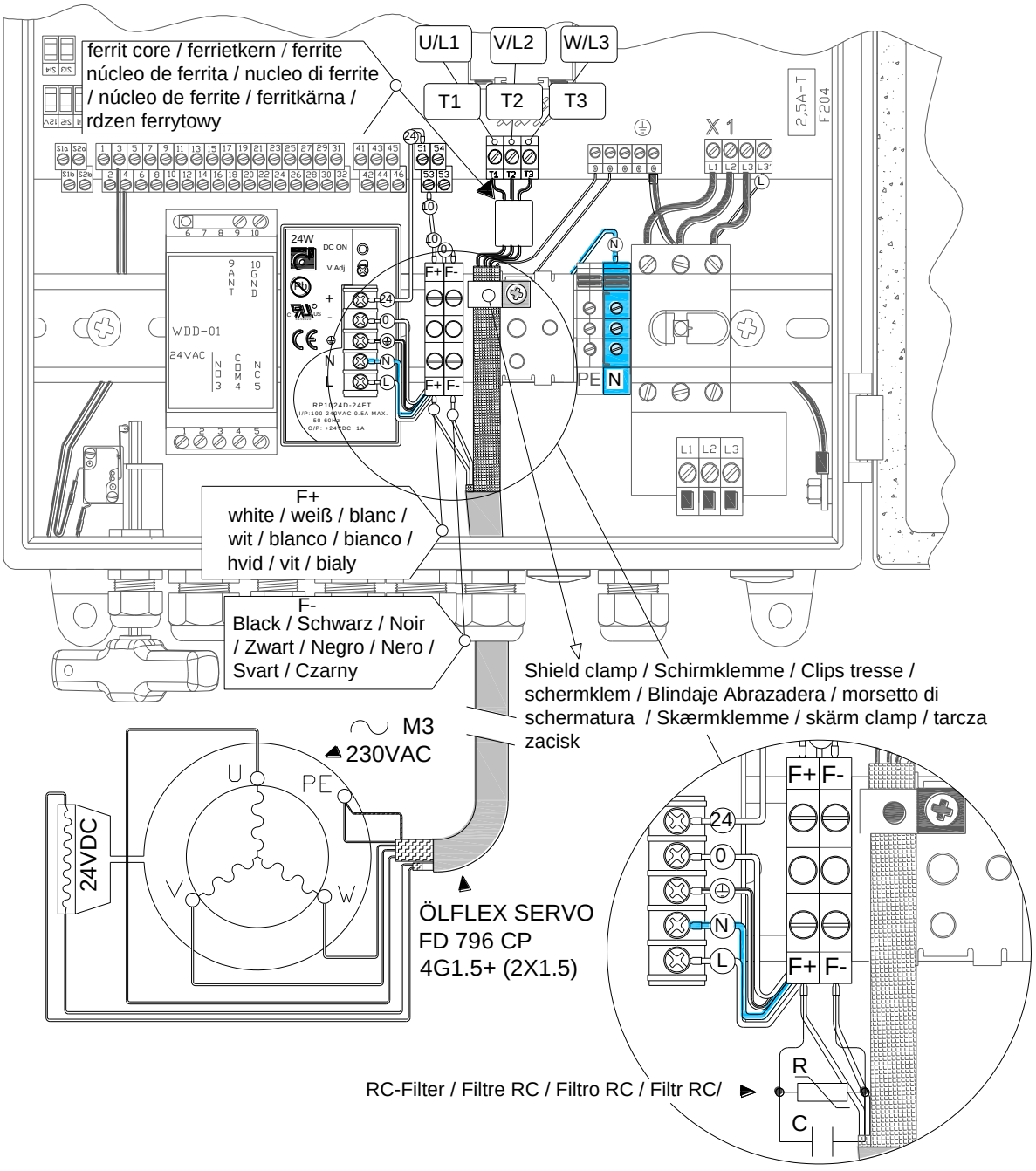


Figure 6-7: Motor connection for door HS9030G and HS9020G three phase 400VAC + neutral.

	To guarantee the trouble-free operation of the 820 DCS controller, a shielded cable must be used. Furthermore, no other wiring may be installed except the wiring for the connection of the motor.

6.2.3. Motor connection for door HS8010P, HS8020P, HS8110P, HS9010P, HS9020G and HS9110P

6.2.3.1. Single phase 230VAC

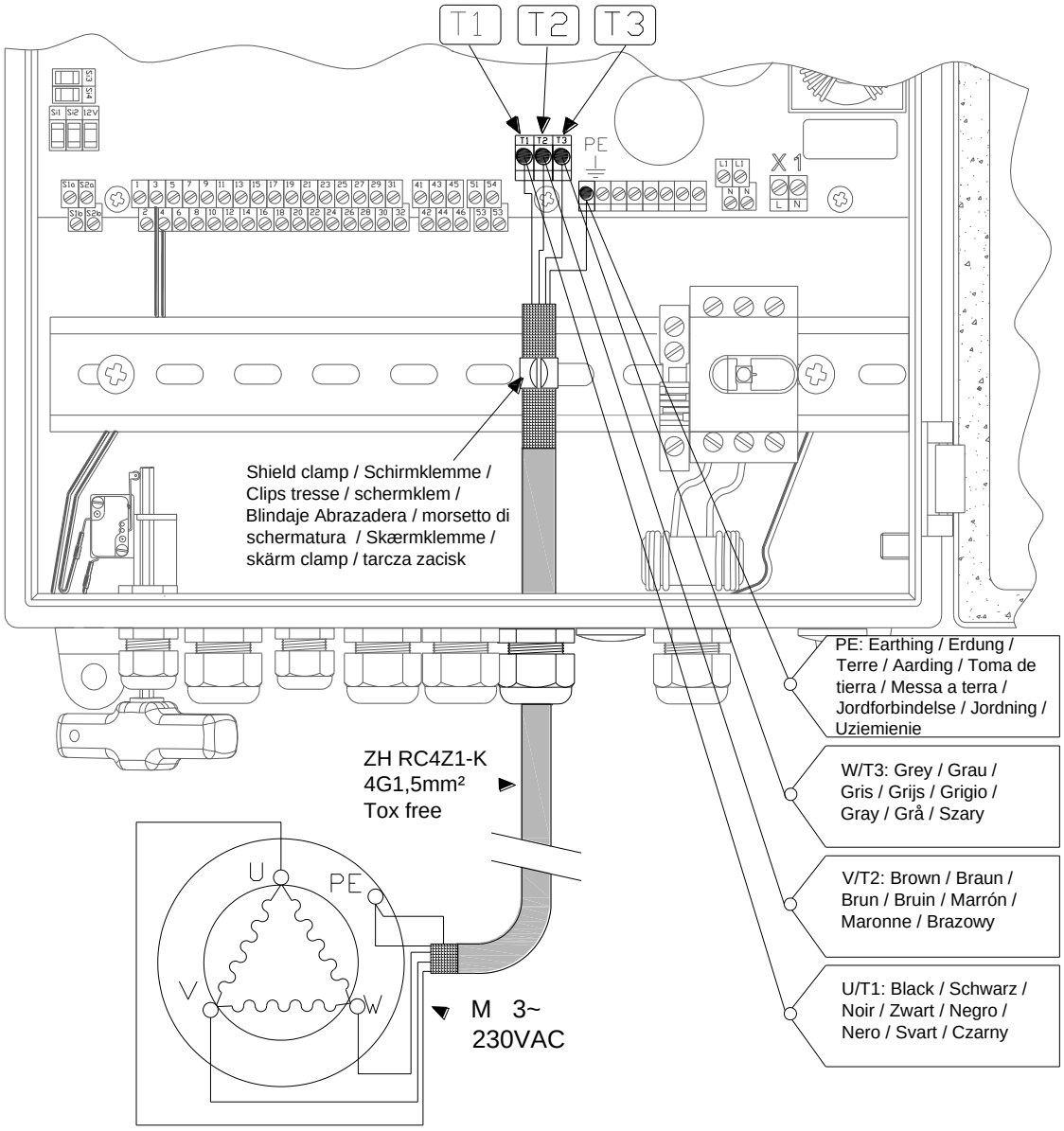


Figure 6-8: Motor connection for door HS8010P, HS8020P, HS8110P, HS9010P, HS9020G and HS9110P – Single phase 230VAC controller

	To guarantee the trouble-free operation of the 820 DCS controller, a shielded cable must be used. Furthermore, no other wiring may be installed except the wiring for the connection of the motor.

6.2.3.2. Three phase 400VAC

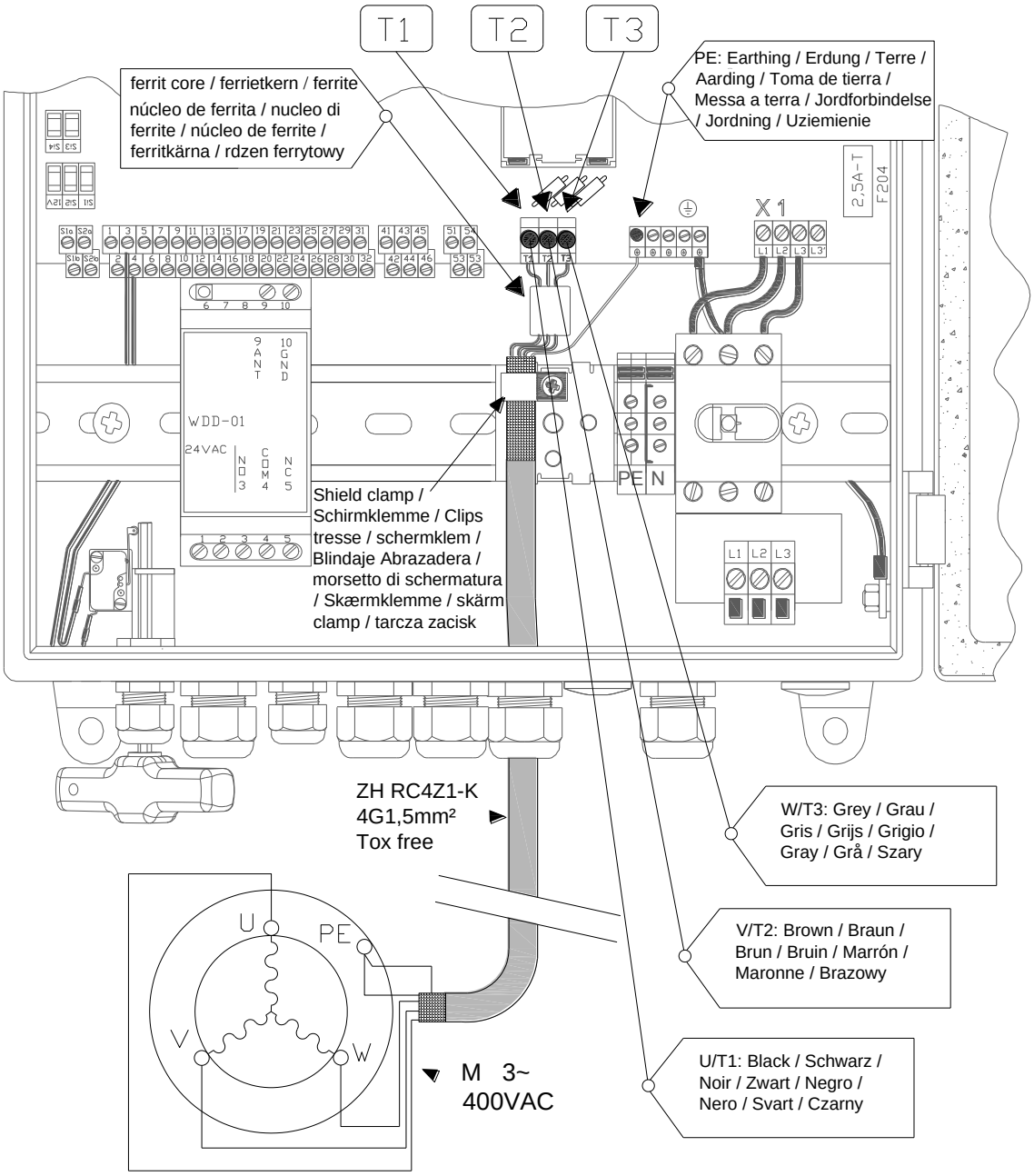


Figure 6-9: Motor connection for door HS8010P, HS8020P, HS9010P, HS9020G, HS8110P and HS9110P – Three phase 400VAC controller

	To guarantee the trouble-free operation of the 820 DCS controller, a shielded cable must be used. Furthermore, no other wiring may be installed except the wiring for the connection of the motor.

6.2.4. Limit switch connection

In the standard version an absolute encoder is used for the limit switch.

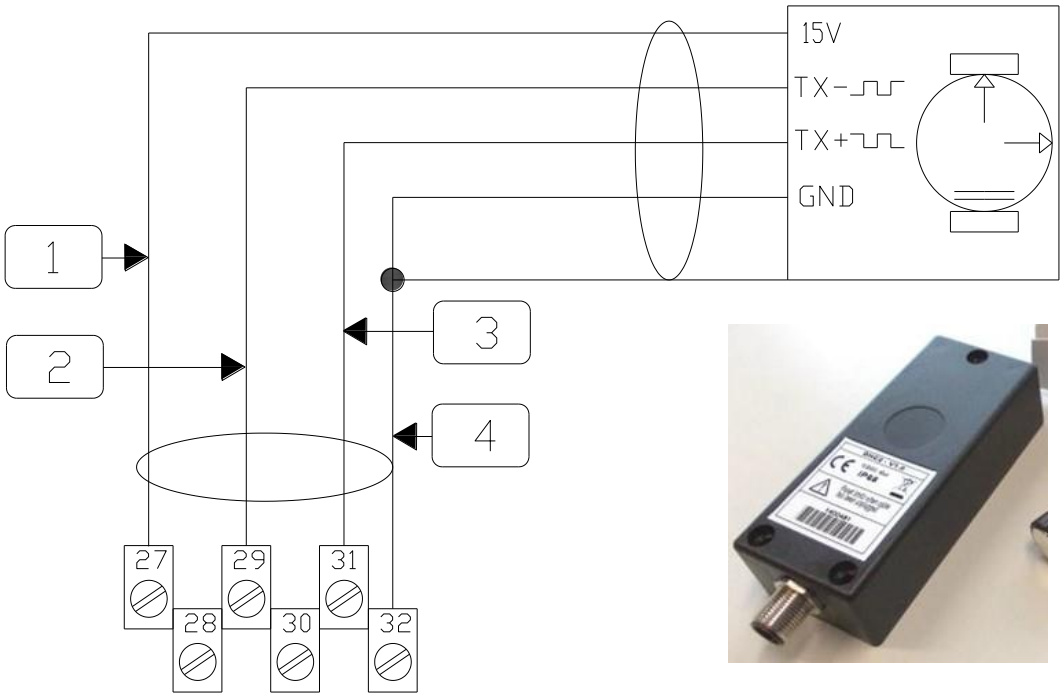


Figure 6-10: Limit switch connection.

1	No. 27 White wire to 12 V.
2	No. 29 Yellow wire to TX-.
3	No. 31 Green wire to TX+.
4	No. 32 Brown wire + shield to GND.
Always available.	

7. Input connections

7.1. Connecting of induction loop(s). (Optional)

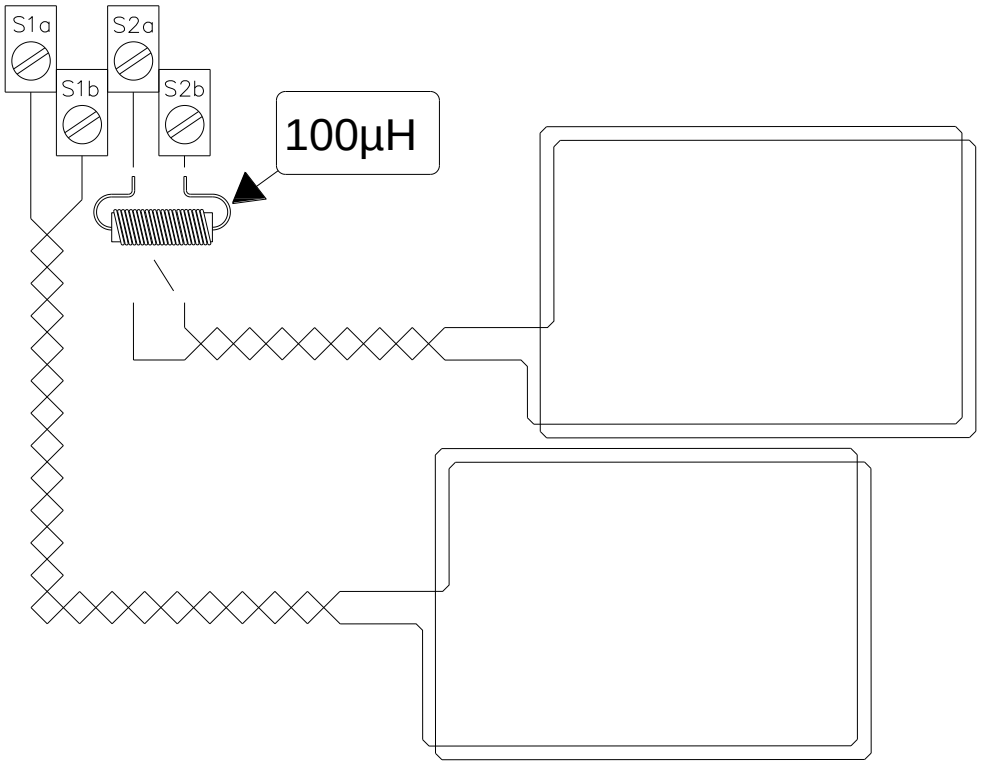


Figure 7-1: Detection loop connection

S1a-S1b: loop 1- Message E501, if loop is activated.

S2a-S2b: loop2.- Message E502, if loop is activated

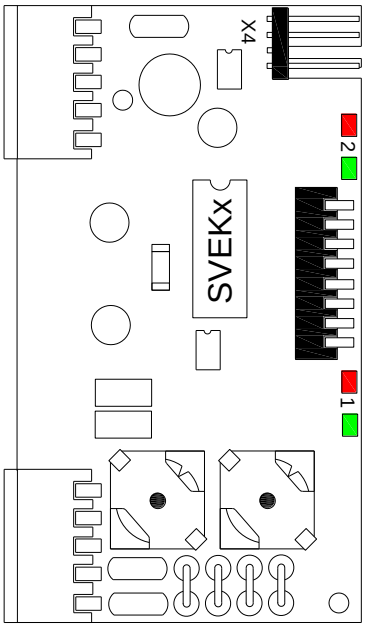
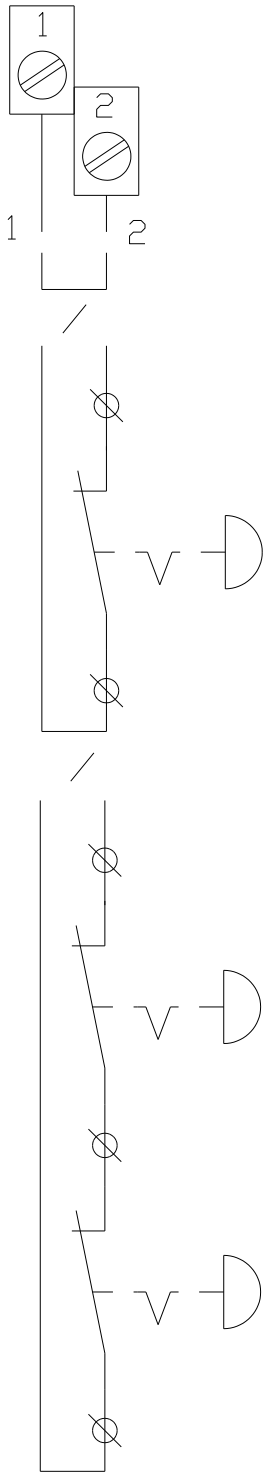


Figure 7-2: Detection loop board

The use of induction loops is possible only if a plug-in card is clipped on the main board. Tuning (for example sensitivity) is also done with this additional card.

7.2. Bridging or connection of an external emergency stop (optional)



Connection is always available.

Material is optional.

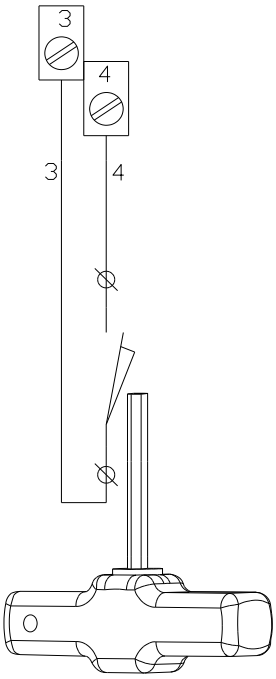
E211 F211 message when the contact is open.

No message when the contact is closed.

Figure 7-3: External E-stop connection.

7.4. Safety contact of the handle for manual maneuver of the door

7.4.1. Crank switch in control box





ATTENTION: If the handle is removed from its support, the contact is open and all functions of the door stop.

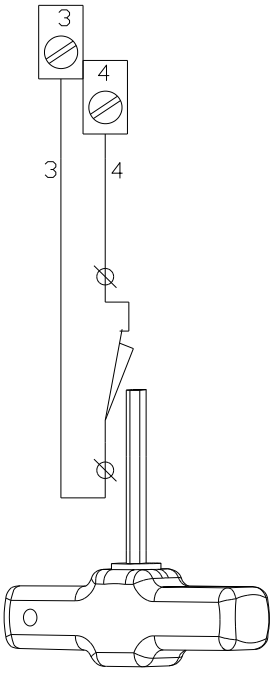
Connection is always available.

Material is always provided.

E212 F212 message when the handle has not been put in the contact.

Figure 7-4: Safety contact of the handle, crank switch in control box.

7.4.2. Crank switch on back side of motor





ATTENTION: If the handle is inserted in the motor, the contact is open and all functions of the door stop.

Connection is always available.

Material is always provided.

E212 F212 message when the handle is inserted in the motor.

Figure 7-5: Safety contact of the handle, crank switch on back side of the motor.

7.5. Opening commands (optional)

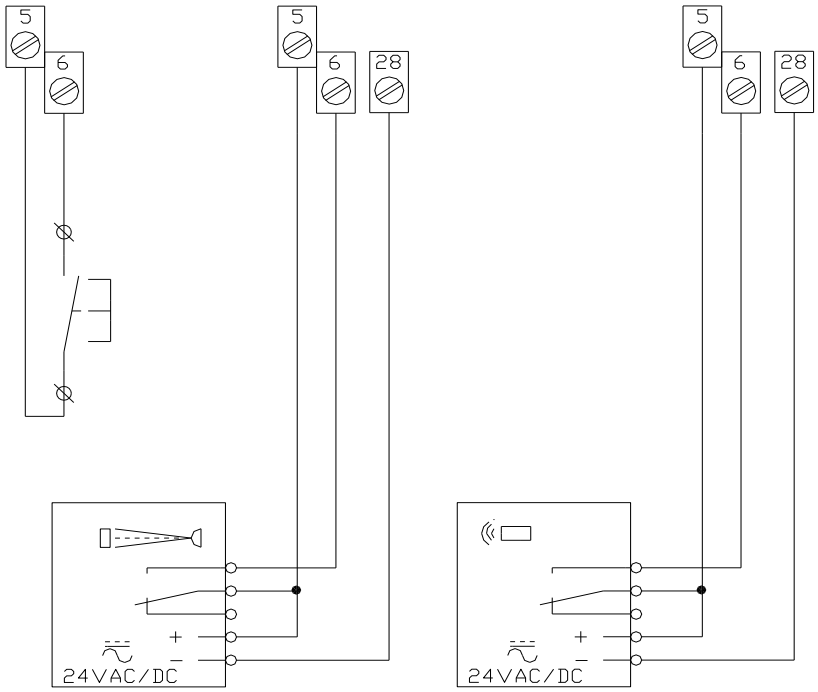


Figure 7-6: Connections of opening commands.

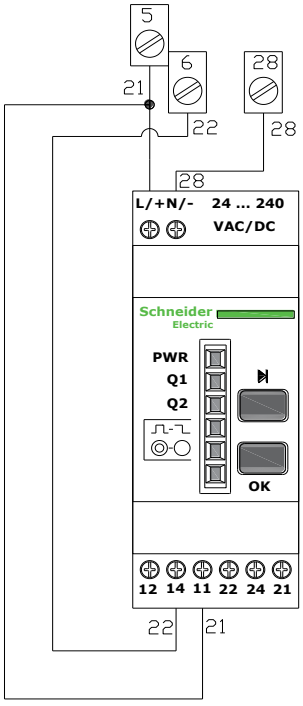


Figure 7-7: Connections of wireless opening commands.

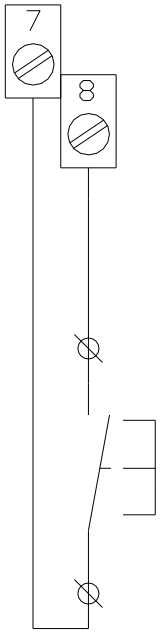
.

Connection is always available.

Material is optional

E101 message when the contact is activated

7.6. Close command (optional)



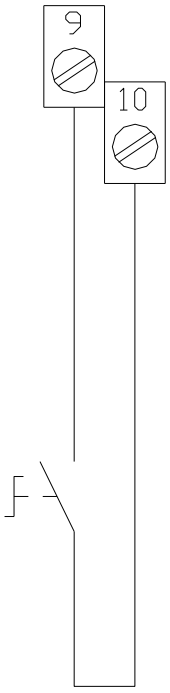
Connection is always available.

Material is optional.

E102 message when the contact is activated

Figure 7-8: Close command connection.

7.7. Auto/manual connection (optional)



ATTENTION: If the contact is open, the system will only function in automatic mode.

Connection is always available.

Material is optional.

E103 message when the selector switch is set to MANU. (Manual).

No message when the selector switch is set to AUTO.

Figure 7-9: Auto/manual connection.

7.9. Infrared devices connection

7.9.1. Photocell

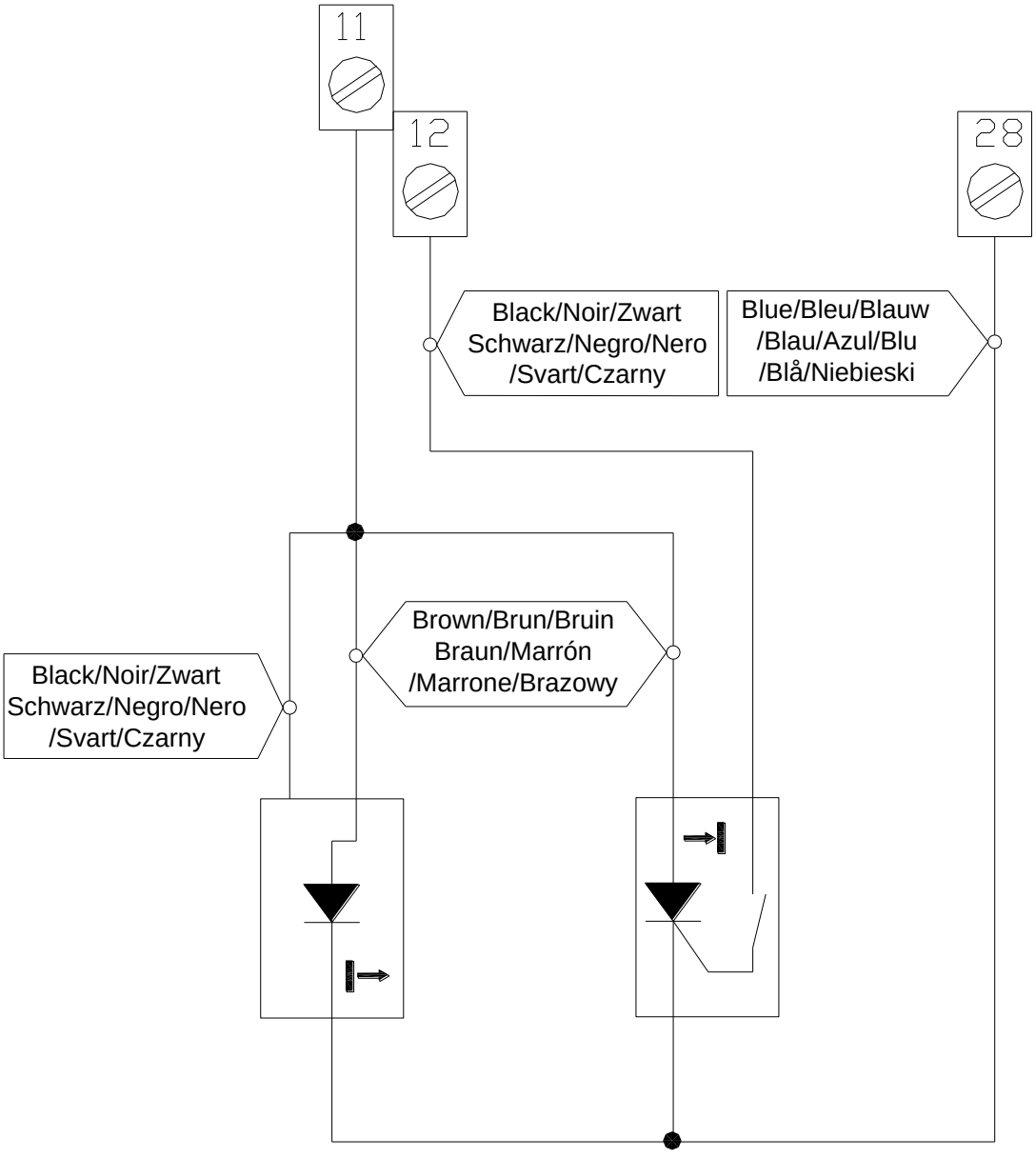


Figure 7-10: Infrared photocell connection.

Always available.

E104 message when the photocell is activated.

7.9.2. Industrial Plus MOVEMENT and PRESENCE detection

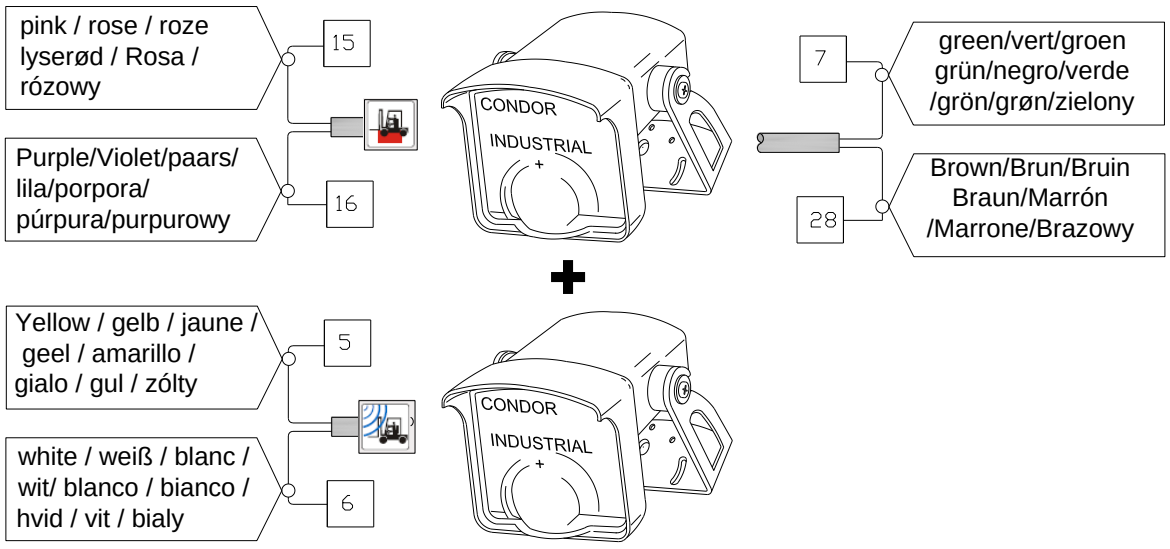
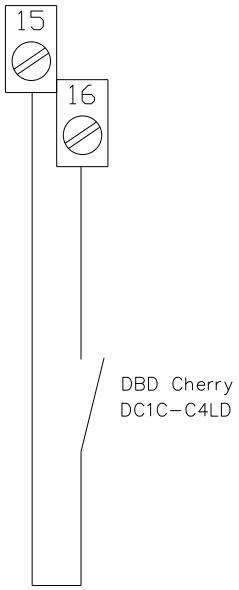


Figure 7-11: Industrial plus movement and presence detection

E106 message when PRESENCE is detected.

E101 message when MOVEMENT is detected.

7.10. Connecting the DBD unwinding detector (doors HS9020G and HS9030G)

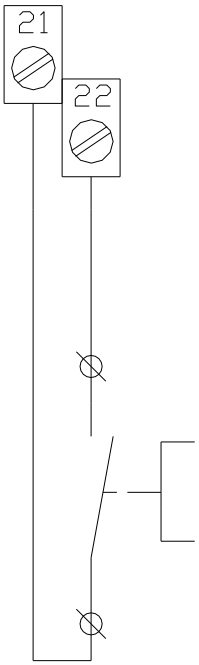


Always available for doors HS9020G and HS9030G.

E106 message when the DBD is active.

Figure 7-12: Connecting the DBD unwinding detector.

7.11. Pull switch connection (optional) – Mid-height opening

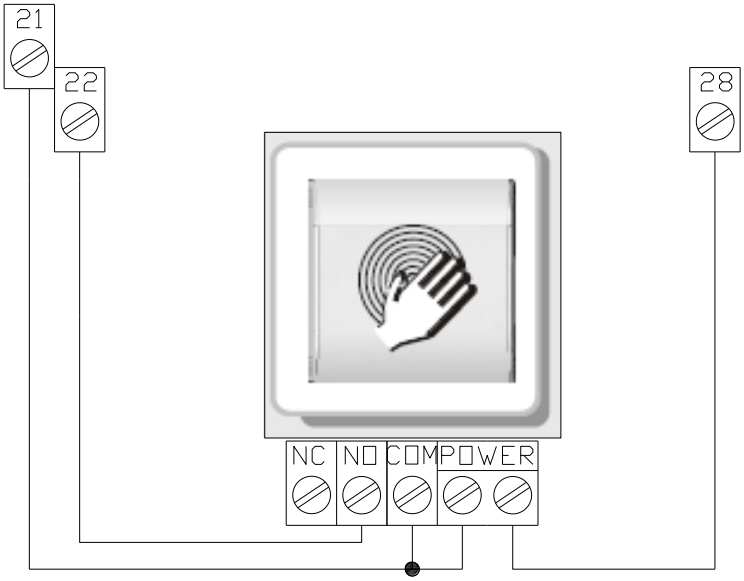


Connection is always available.

Material is optional.

E109 message when pull switch is activated.

Figure 7-13: Pull switch connection.



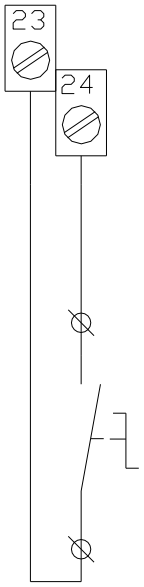
Connection is always available.

Material is optional.

E109 message when Magic switch is activated.

Figure 7-14: Magic switch as mid opening connection.

7.13. Deactivation of a function (optional)



Connection is always available.

Material is optional.

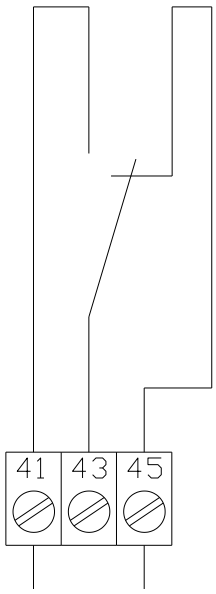
E110 message when the selector switch is set to ON.

No message when the selector switch is set to OFF.

Figure 7-15: Deactivation of a function connection.

8. Output connections

8.1. Door without traffic lights: “door open” information

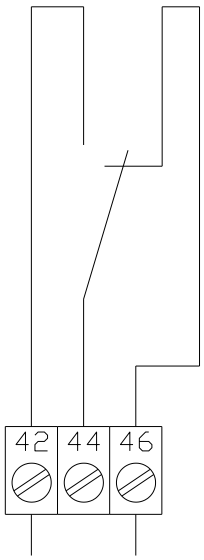


Always available

The relay switches when the door reaches the upper limit switch.
Terminal 43 is common

Figure 8-1: Door without traffic lights: “door open” information.

8.2. Door without traffic lights: “door closed” information



Always available

The relay switches when the door reaches the lower limit switch.
Terminal 44 is common

Figure 8-2: Door without traffic lights: “door closed” information..

8.4. Connection optional flashing light

- 24V Orange flashing light for Door HS9030G and HS9020G without pre-warning (Only available with single phase 240VAC Controller or Three phase 400VAC+Neutral)

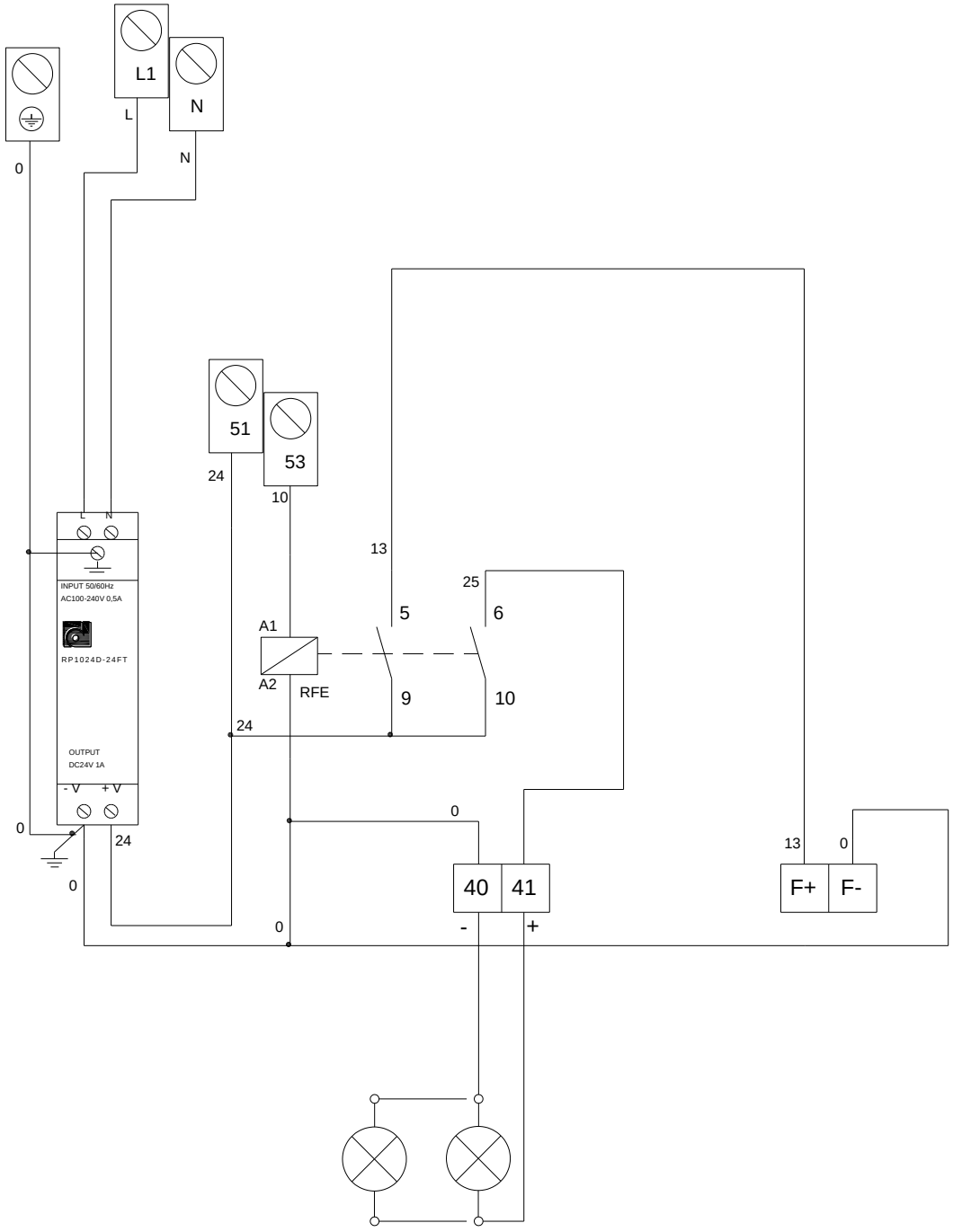


Figure 8-3: 24V connection for door HS9030G and HS9020G without pre-warning.

Connection is always available.

Material in option

- 24V Orange flashing light for Door HS8010P, HS8020P, HS8110P, HS9010P, HS9020G and HS9110P without pre-warning.

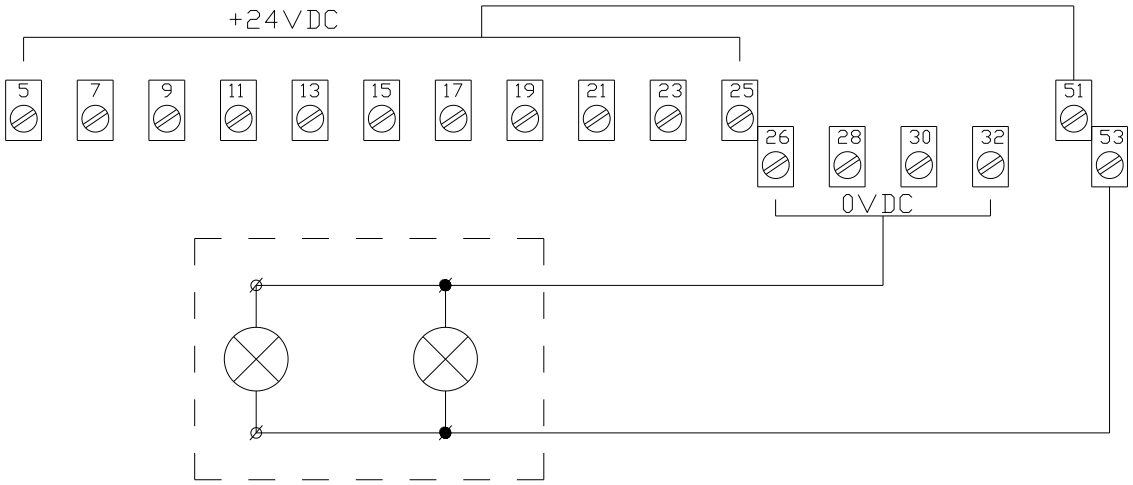


Figure 8-4: 24 V Orange flashing light for door HS8010P, HS8020P, HS8110P, HS9010P, HS9020G and HS9110P.

Connection is always available. Material is optional.

- 24V Orange flashing light with pre-warning for Door HS8010P, HS8020P, HS8110P, HS9010P, HS9020G HS, HS9030G and HS9110P. (Optional)

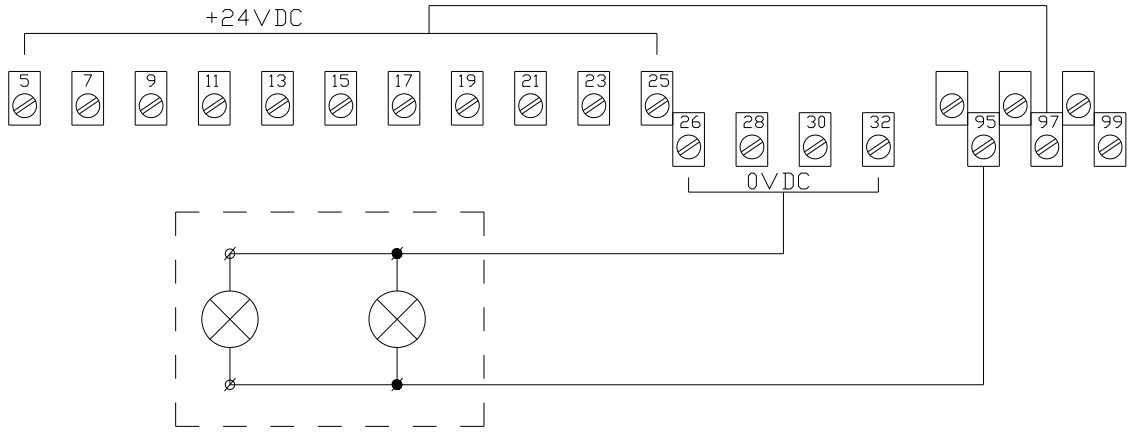


Figure 8-5: 24V Orange flashing light with pre-warning for Door HS8010P, HS8020P, HS8110P, HS9010P, HS9020G, HS9030G and HS9110P.

- 230V Orange flashing light for door HS8010P, HS8020P, HS8110P, HS9010P, HS9020G and HS9110P without pre-warning (Only available with single phase 240VAC Controller)

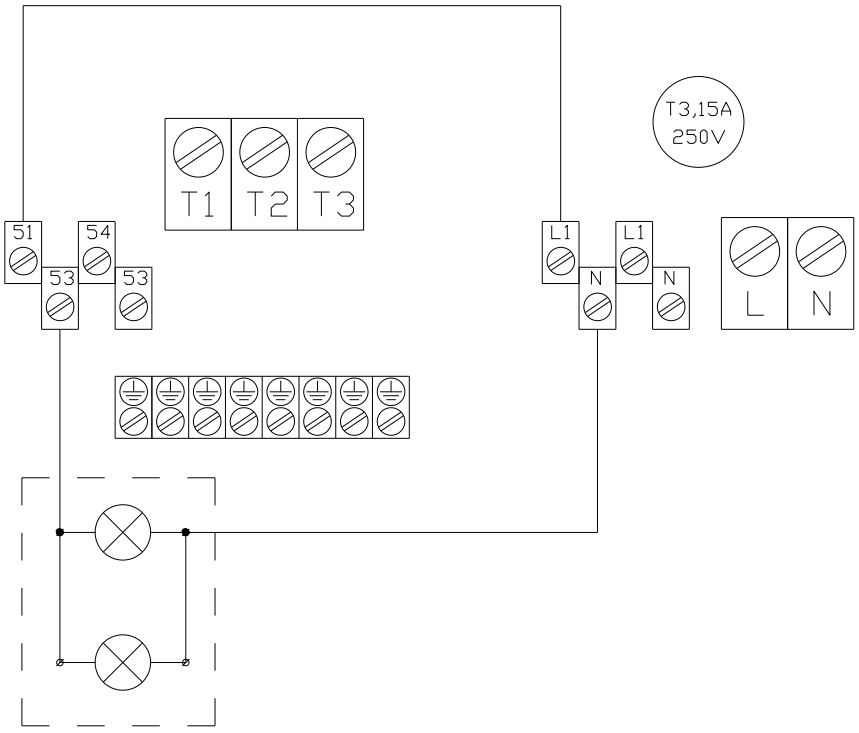


Figure 8-6: 230V Orange flashing light for door HS8010P, HS8020P, HS8110P, HS9010P, HS9020G and HS9110P without pre-warning.

Attention: 3.15 A fuse	
Connection is always available.	Material in option

- 230V Orange flashing light with pre-warning (Only available with single phase 240VAC Controller)

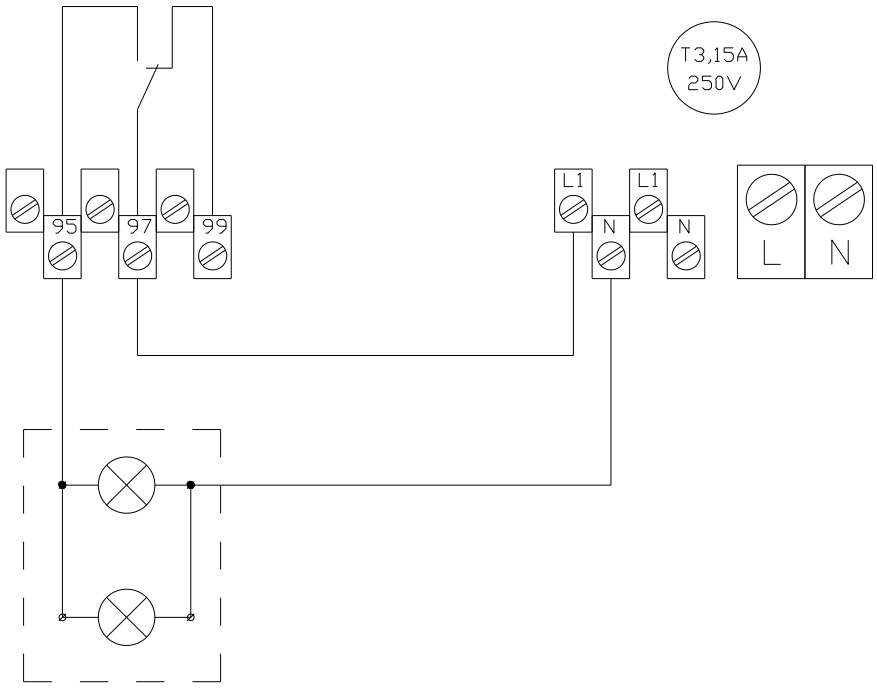
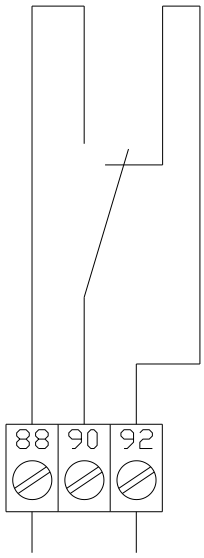


Figure 8-7: 230V flashing light with pre-warning.

Attention: 3.15 A fuse

8.5. Door without traffic lights: “door out of service” information



Optional.

The relay switches when the door is out of service.
Terminal 90 is common

Figure 8-8: Door without traffic lights: “door out of service” information.

8.6. Door with 240VAC traffic lights: connecting the traffic lights

Max.: 600 Watts.

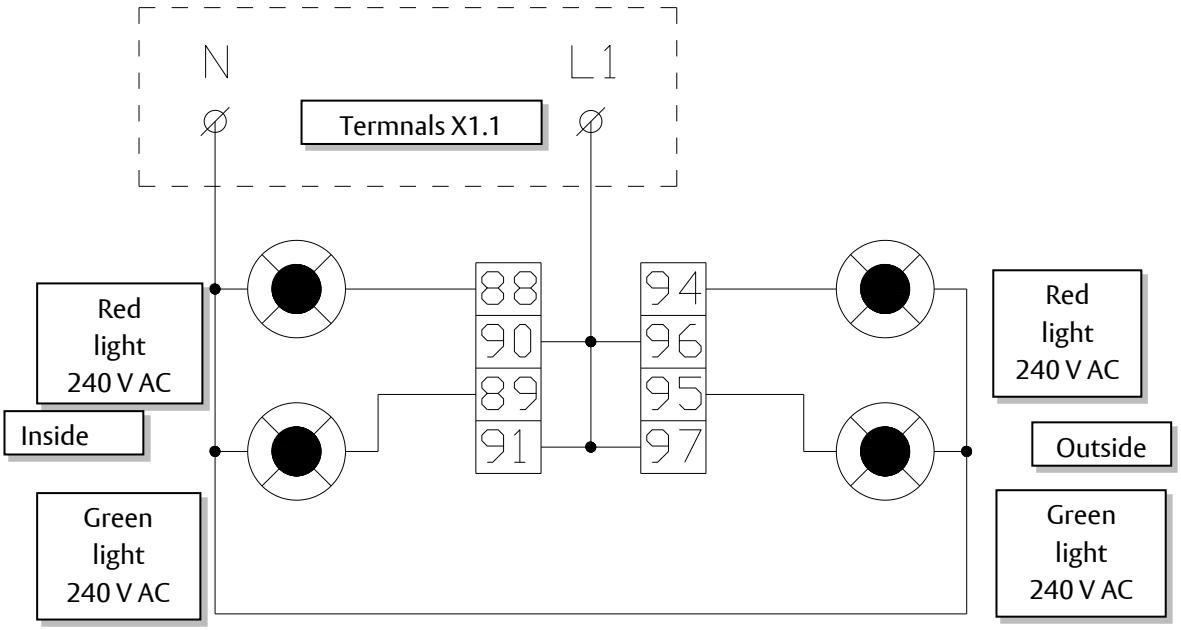
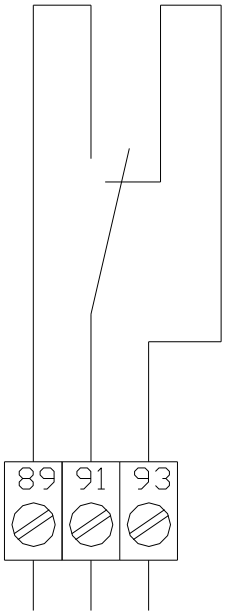


Figure 8-9: Door with traffic lights: connecting the traffic lights.

8.7. Door with traffic lights: “door open” information

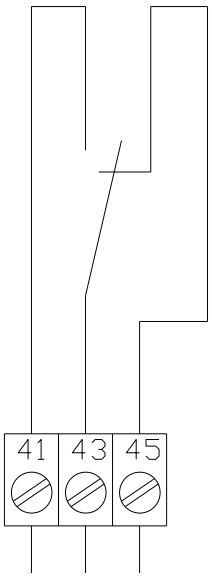


Always available.

Figure 8-10: Door with traffic lights: door open

Because the traffic lights already use too many relays, there are none available for this specific option. Output 7 (connection terminals 89 91 93) can also be used to maintain this information as long as the operation of the relay is not changed.

8.8. Door with traffic lights: “door closed” information



Always available

Figure 8-11: Door with traffic lights: "door closed" information

Because the traffic lights already use too many relays, there are none available for this specific option. Output 5 (connection terminals 41 43 45) can also be used to maintain this information as long as the operation of the relay is not changed.

9. Airlock option

9.1. Door without traffic lights: connection for AIRLOCK function

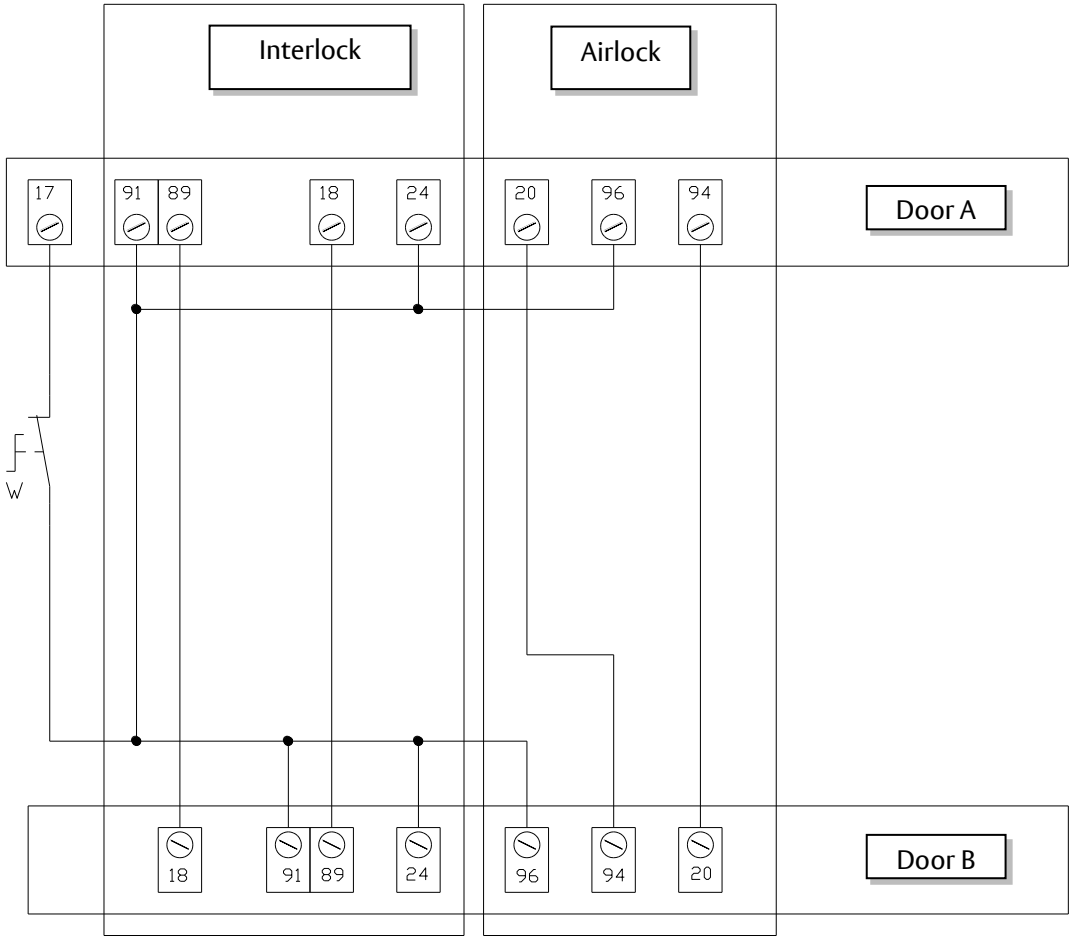


Figure 9-1: Door without traffic lights: connection for AIRLOCK-function.

Door A, message E107 (connection terminal 18): Door A is locked in CLOSED position.

Door A, message E108 (connection terminal 20): Door A receives an open command from door B

Door A, message I130 (connection terminal 94 / 96): Door A sends an open command to door B.

Connection terminals 89/91: Door A locks door B in the CLOSED position.

W: AIRLOCK ON/OFF switch

9.3. Door with traffic lights: connection for AIRLOCK-function

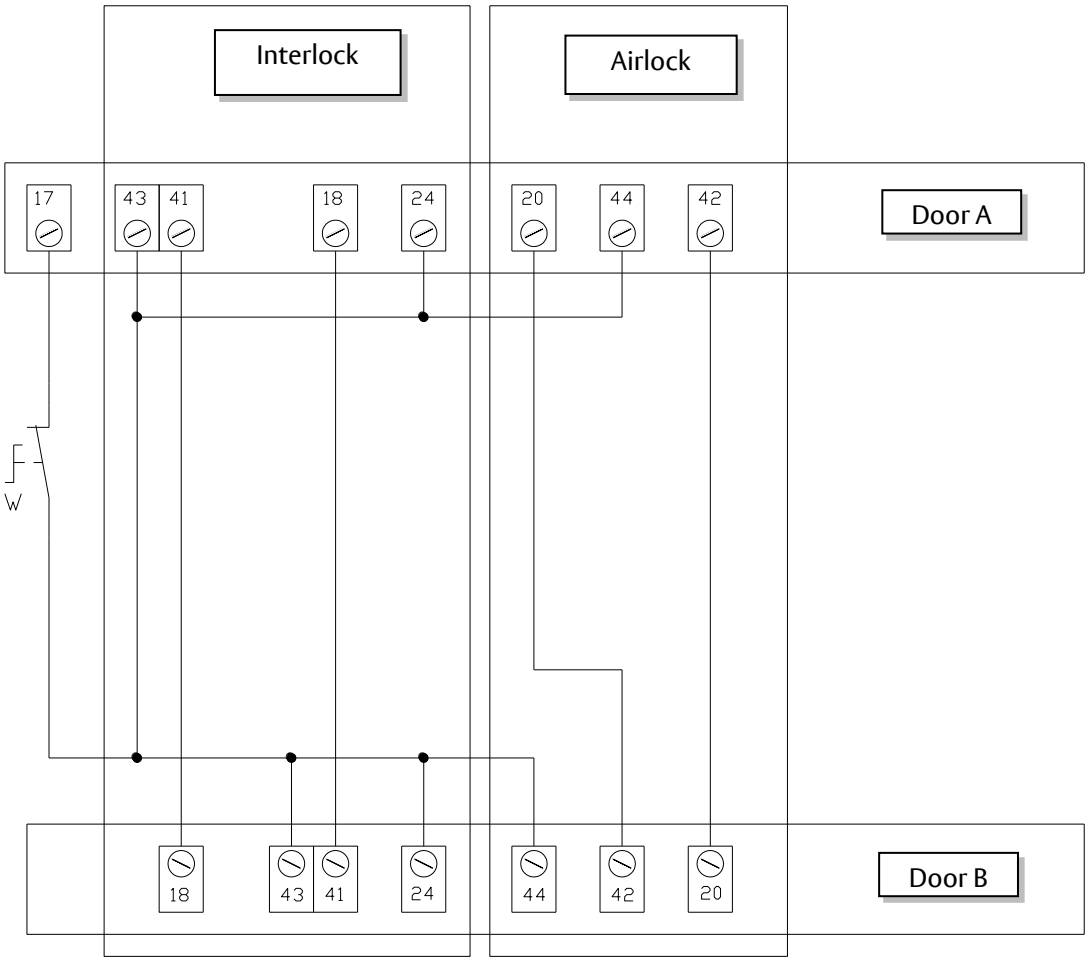


Figure 9-2: Door with traffic lights: connection for AIRLOCK-function.

Door A, message E107 (connection terminal 18): Door A is locked in the CLOSED position.

Door A, message E108 (connection terminal 20): Door A receives an open command from door B

Door A, message I130 (connection terminal 42 / 44): Door A sends an open command to door B.

Connection terminals 41/43: Door A locks door B in the CLOSED position.

W: AIRLOCK ON/OFF switch

10. Freezer

10.1. HS9010PFR heating cable connection (Only available with Single Phase 240VAC and Three phase 400VAC + Neutral)

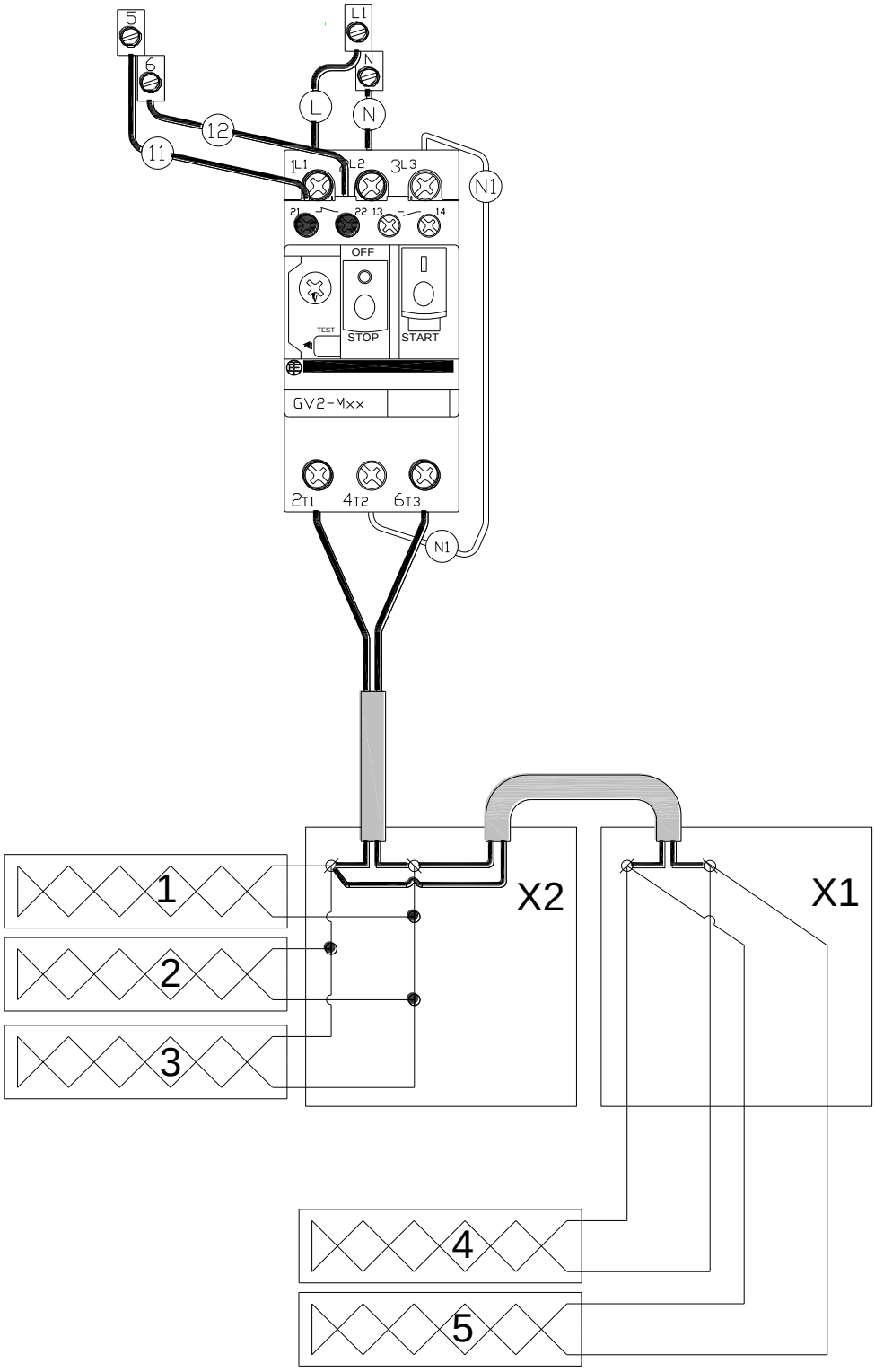


Figure 10-1: H9010PFR heating cable connection.

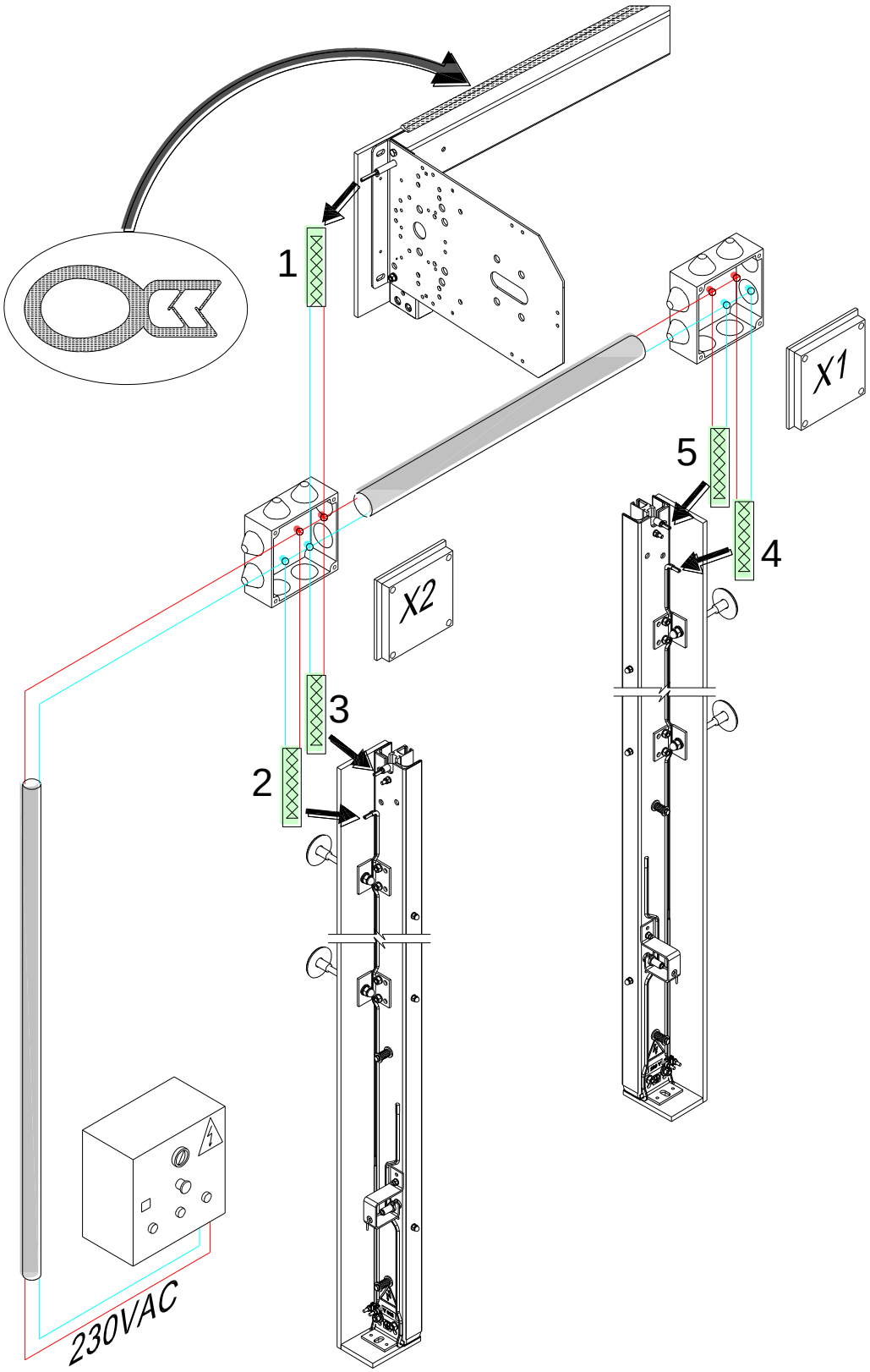


Figure 10-2: HS9010FPR heating cable connection-2.

11. Emergency exit option

11.1. Connection Emergency exit (2 in 1)

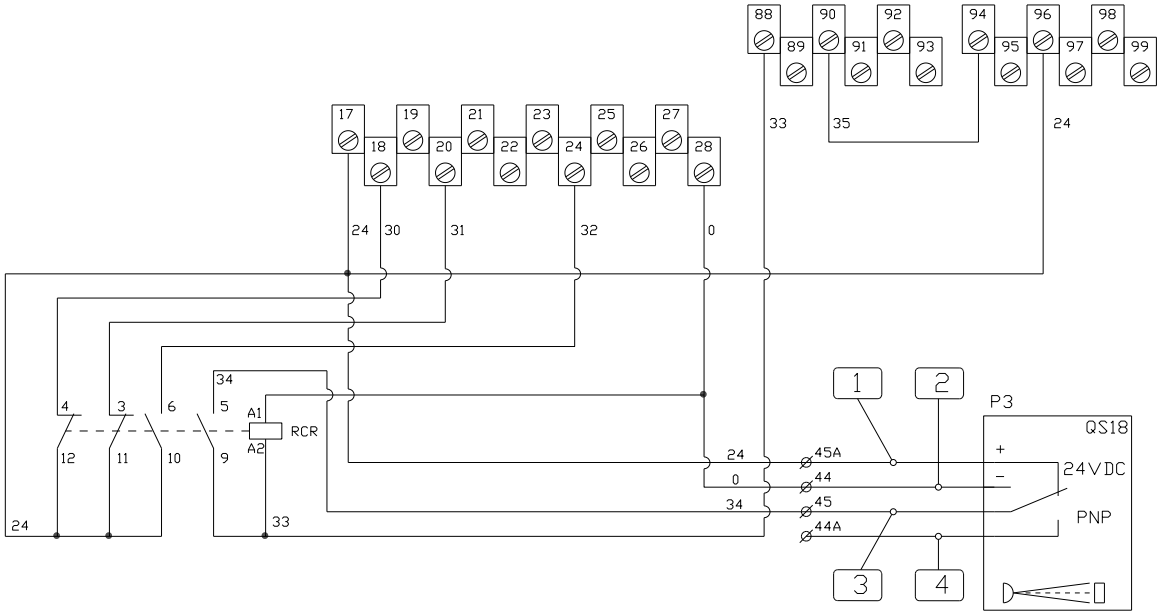


Figure 11-1: Connection Emergency exit.

1: Brown
2: Blue
3: White
4: Black

When the REFLEX Cell sees its reflector, it means the emergency exit is in use. The door stops and can only be used manually. If the door has to be closed, the T-cut should be repaired. (push the emergency stop for your safety)
When the door is repaired, it should be reinitialised:

- Push the emergency stop
- Release the emergency stop
- Press the button ▲ before ten seconds are expired.

Never leave the T-cut open, the door might be damaged.

12. UPS (Option)

12.1. Connecting the UPS. (Only available with Single phase 240VAC Controller)

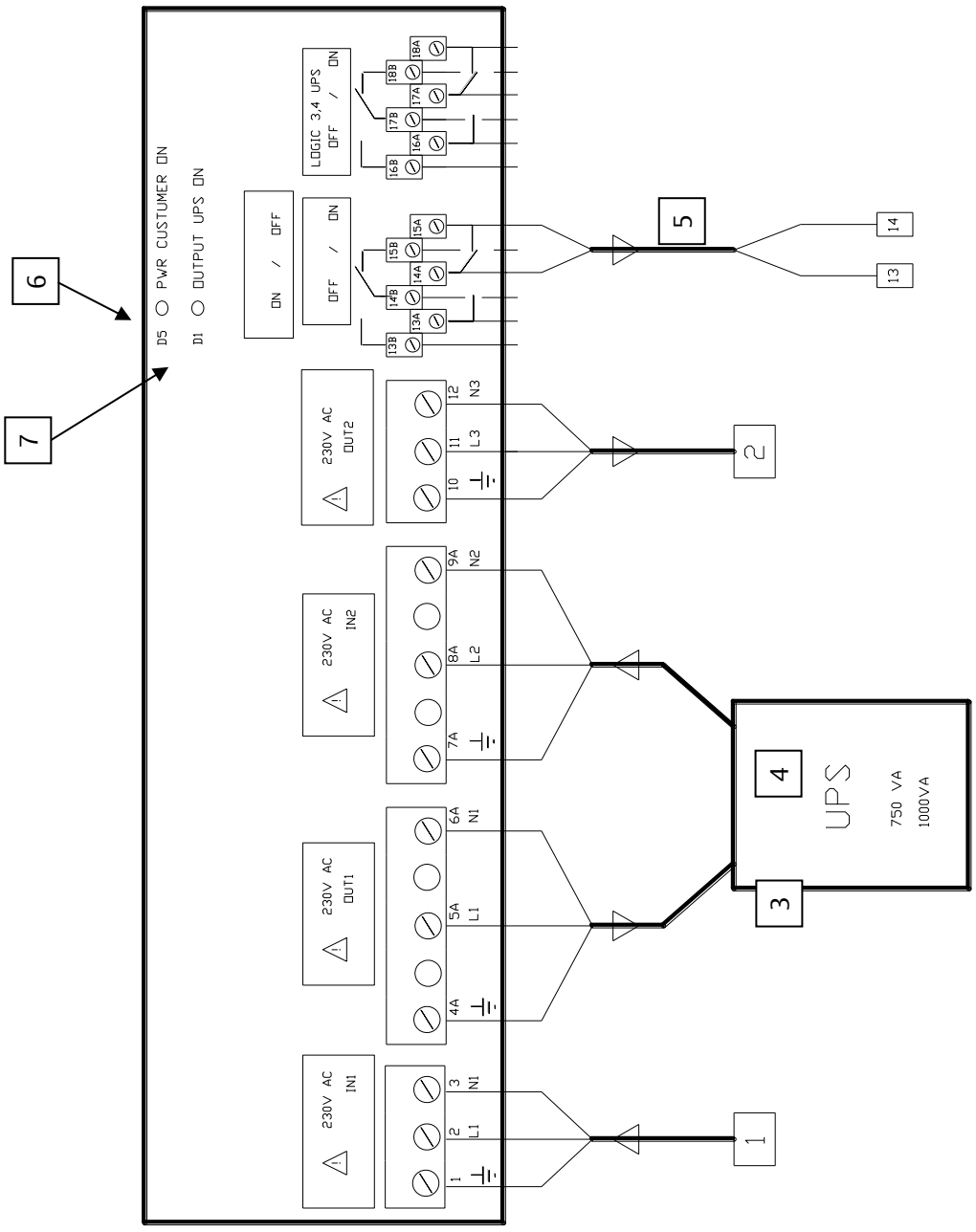


Figure 12-1: Connection of the UPS

1: Power from the mains	4: Emergency power supply from the UPS
2: Power supply to 820 DCS control panel	5: Opening command
3: Power supply to the UPS	△: Direction of power supply
6: D5 green = mains power supply is available	
7: D1 red = output power supply from the UPS is available	
Both are lit under normal circumstances	

13. Wireless ASSA ABLOY detector

13.1. Connection of WDD

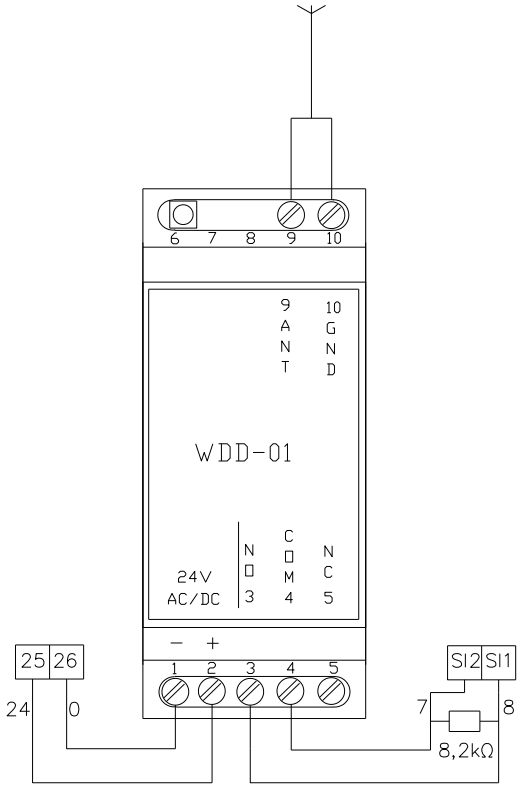
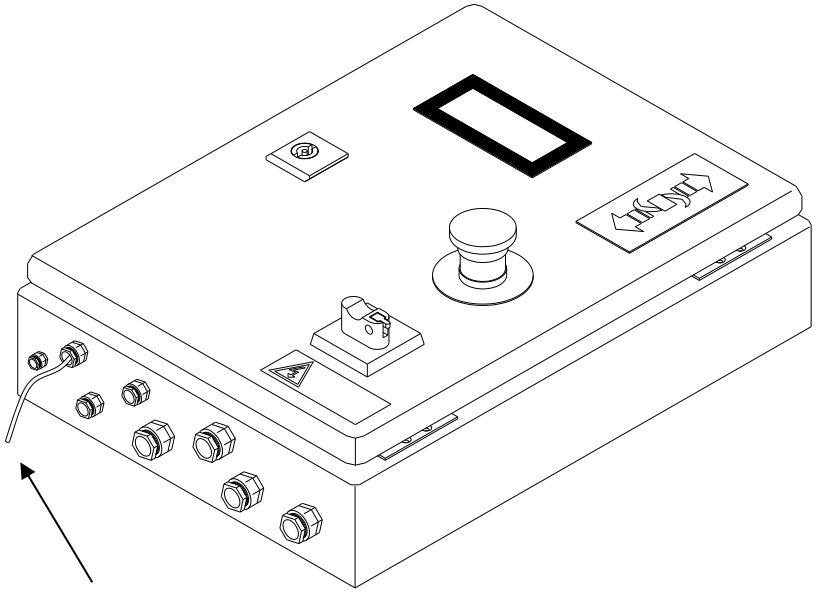


Figure 13-1: Connection of WDD



Antenna:
Pre-cabled on clamps 9 & 10, cable passage PG7 at the bottom of the panel.

Figure 13-2: Antenna of WDD

13.2. Operation

The wireless ASSA ABLOY detector is a wireless system with a sensor. A conductive rubber profile located in the bottom bag and over the whole width of the door is connected to a transmitter. When the sensor hits an obstacle, the transmitter will leave the waiting mode and it will sends a signal to the receiver in the control box, which will immediately reopen the door.

13.3. Adjustment procedure

Programming:
Apply current to the receiver: the L2 red light will come on.
Press the S1 button at the same time put pressure on the sensor rod: the green light will flash to indicate that the code has been registered correctly.
Release S1: the L1 green light will go off.
The programming is OK.

Test: Press the sensor rod; the green light will come on. OK

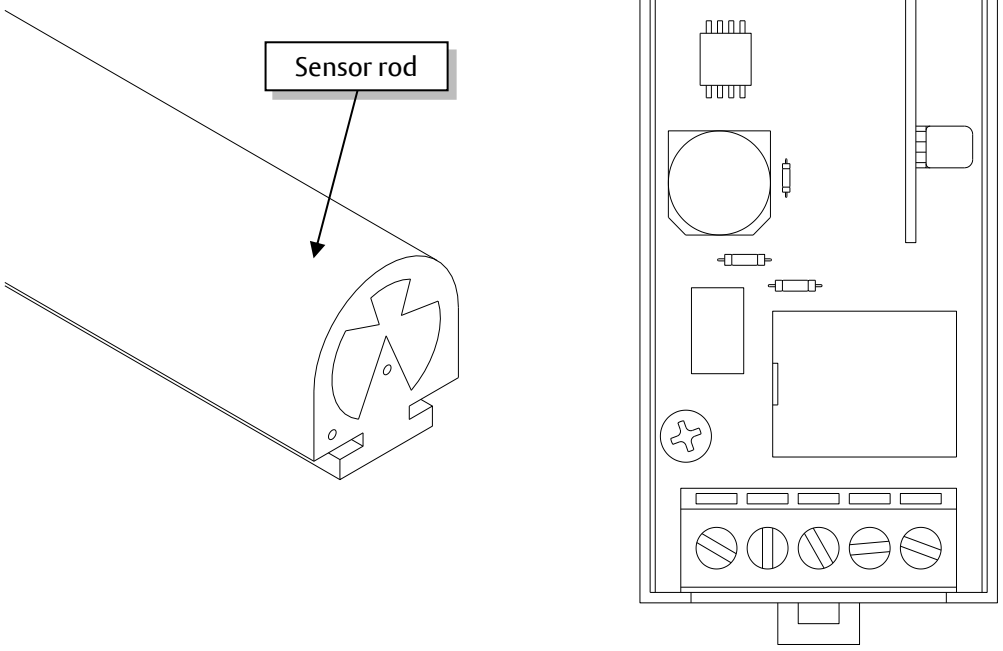
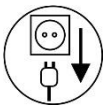
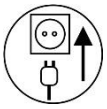
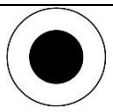


Figure 13-1: Adjustment procedure.

14. Commissioning

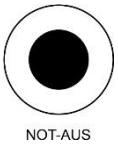
14.1. General procedures for setting parameters

Opening the use of parameters					
1		Switch off the control box.	Switch off the provision for all poles (Please check the safety instructions.)	The display of the 7 segments will slowly switch off after a few seconds.	
2		Switch the DIP switch to ON.	The service mode is actuated. Close the control box afterwards		
3		Switching on the control box.	Switching on the control.	The front decimal point will flash when a service mode has been activated The display of the content will depend on the control status.	.*...
4		Press emergency STOP.	Immediate switch of the door to the use of parameters.	The parameter mode can be immediately activated through the Emergency OFF button.	P*000

Parameter choice when the use of parameters is open					
		OPEN Or CLOSE	Select the required parameter. **ATTENTION:** Not all parameters can be immediately viewed or changed depending on the password and the set position type.	The parameter value can be viewed or changed (see below). The display will differ per selection.	**P....**

Parameter processing of the selected parameter					
1			Control in the parameter mode.	Display of the required parameter name.	P010
2		STOP (short).	Opening the parameter.	The current parameter value will now be displayed.	5
3		Open	The Open button to increase the parameter value.	The current applicable parameter value is being changed; the decimal points are flashing.	6*
Or		Close	The Close button to decrease the parameter value.		4*

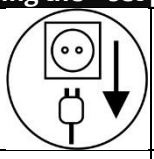
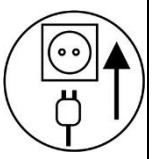
Parameter processing of the selected parameter					
4		STOP (long).	Save the set parameter value.	The parameter value has been saved when no more points are flashing.	6
Or		STOP (short).	Delete the set parameter value	Cancel; the original parameter value will again be displayed.	5
5		STOP (short).	Change the display of the parameter name.	Display of the parameter name.	P 0 1 0

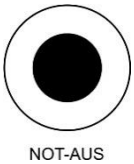
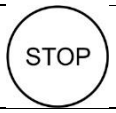
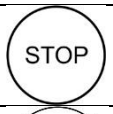
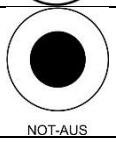
Complete the use of parameters.					
		Release Emergency STOP (pull).	The use of parameters will be immediately terminated; door use will again be activated.	The use of parameters can always be terminated by using Emergency OFF. The last saved value will remain automatically saved.	

	After approximately 1 hour the use of services will be independently restored. The control must be deactivated for a short period of time and then again activated or a restore must be performed to return to the use of services.

14.2. Basic settings

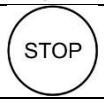
14.2.1. New settings of the limit switches

Opening the "'set parameters" function								
1		Switch off the door control.	Switch off the power supply to all poles (please check the safety instructions)	The display of the 7 segments will go out after a few seconds.				
2		Switch the DIP switch to ON.	The service mode is activated. Close the control box afterwards.					
3		Activating the door control.	Activating controls.	The first decimal point will flash when a service mode has been activated. Display of the content will depend on the control status.	.*	.	.	.

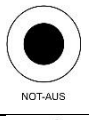
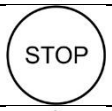
Opening the "'set parameters" function								
4		Press emergency STOP button.	Immediate switching over of the door to the use of parameters	The parameter mode can be activated immediately through the Emergency OFF button.	P*	0	0	0
5		OPEN or CLOSE	Select the required parameters PLEASE NOTE: Not all parameters can be viewed/changed immediately.	The value of the parameters can be viewed or changed. The display will differ in accordance with the selection.	P.	2	1	0
6		STOP (short).	Opening of the parameter	The current value of the parameter will be indicated				0
7		OPEN.	To increase the value.	The decimal points will flash if the currently applicable value is changed				3 *
8		STOP (long).	Save the value that has been set.	The parameter has been saved when all points have stopped flashing.				3
9		STOP (short).	Back to the parameter	Indication of the name of the parameter.	P.	2	1	0
10		Emergency stop.	Pull OUT the "Emergency stop" button.	Access to the "'calibration" mode.	C.	A	L	i
11		STOP (short).		The indication is asking to search for the end point for the closed condition.	**E.**	**I.**	**E**	**U.**
12		OPEN or CLOSE	Limit switch "door closed"	As long as the end position of closed condition has not been validated, the decimal points will continue to flash.	**E.**	**I.**	**E**	**U.**
If the direction of rotation of the motor is incorrect, go to parameter P.130 and change the direction of motor rotation (from 0 to1). Next, follow this procedure								
13		STOP (long).	Save the "closed" position.	The parameter has been saved when all points have stopped flashing.	**E.**	**I.**	**E**	**U.**

Opening the "'set parameters" function								
14			The system will automatically switch over to calibration of the limit switch "door open".	The indication is asking to search for the end position of the opening travel.	E.	I.	E	O.
15		OPEN Or CLOSE	Adjust limit switch "door open".	As long as the end point for the opening movement has not been validated, the decimal points will continue to flash.	E.	I.	E	O.
16		STOP (long).	Save the "open" position.	The parameter has been saved when all points have stopped flashing.	E.	I.	E	O.
			The system will automatically switch to the intermediate stop setting/2nd opening height.	The indication requires the intermediate stop position to be set.	E.	I.	E	I.
17		OPEN or CLOSE	Set the 2nd opening height, EVEN WHEN YOU ARE NOT GOING TO USE IT.	The values on the screen will flash until the end position is saved.	E.	I.	E	I.
18		STOP (long)	Save the 2nd opening height.	The input will only be finished when all symbols on the screen have stopped flashing.	E.	I.	E	I.
			The door is ready for use.		, ,			, ,
19			Allow the door to come down to check the limit switch door closed.					
20			Activate the open command to check the limit switch door open.					

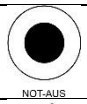
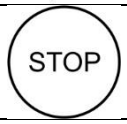
14.2.2. Fine adjustment of limit switch door closed

1		Press emergency STOP button.	Open the "set parameters" mode.		**P.**	**0**	**0**	**0**
2		OPEN or CLOSE	Select the P.221 parameter.	The parameter will be displayed.	**P.**	**2**	**2**	**1**
3		STOP (short)	Show the value of the parameter.	The default value will be displayed.				**0**
4		OPEN or CLOSE	A positive value will set the position of the limit switch higher. A negative value will set it lower. Values are given in increments (=±5mm)	We advise, not to set this value to high or to low at the first time, you may go beyond the desired limit switch.			**-**	**3 ***
5		STOP (long).	Save the corrected "closed" position.	The parameter has been saved when all points have stopped flashing.				**3**
6		Release the Emergency STOP button.	Exit the "set parameters" mode.	Open the door and observe whether the limit switch is correct. If necessary, start again from line 1.				

14.2.3. Fine adjustment of limit switch door opened

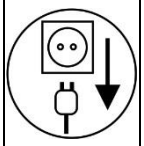
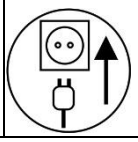
1		Press emergency STOP button.	Opening of the “set parameters” mode.		**P.**	**0**	**0**	**0**
2		OPEN or CLOSE	Select the P.231 parameter.	The parameter will be displayed.	**P.**	**2**	**3**	**1**
3		STOP (short).	Show the value of the parameter.	The default value will be displayed.				**0**
4		OPEN or CLOSE	A positive value will set the position of the limit switch higher. A negative value will set it lower. Values are given in increments.	We advise, not to set this value to high or to low at the first time, you may go beyond the desired limit switch.				**3 ***
5		STOP (long).	Save the corrected “open” position.	The parameter has been saved when all points have stopped flashing.				**3**
6		Release the Emergency STOP button.	Exit the “set parameters” mode.	Open the door.				

14.2.4. Setting the timing for the closing cycle

1		Press emergency STOP button.	Opening of the "set parameters" mode.		**P.**	**0**	**0**	**0**
2		OPEN or CLOSE	Select the P.010 parameter.	The parameter will be displayed.	**P.**	**0**	**1**	**0**
3		STOP (short).	Show the value of the parameter.	The default value will be displayed.				**7**
4		Open.	Increase the time	The decimal points will flash if the currently applicable value is changed.				**9***
5		CLOSE.	Decrease the time	The decimal points will flash if the currently applicable value is changed.				**4***
6		STOP (long).	Save the value (in seconds) of the timing.	The parameter has been saved when all points have stopped flashing.				**4**
7		STOP (short).	Cancels the "set parameters" mode.	The default value will again be displayed.				**7**
8		Release the Emergency STOP button.	Exit the "set parameters" mode.	Open the door.	**T**			**7**

14.2.5. Exiting the “service mode”

	Before finally putting the door into operation, you must first exit the “service mode”.

1		Disconnect the power supply from the control panel.	Disconnect the power supply from all external and internal command circuits; please observe the safety rules	The 7-segment display will switch off after a few seconds.				
2		Switch the DIP switch to OFF.	The “service” function will be deactivated; close the electrical controlbox.	(Position S1, see section 6.3)				
3		Power up the control panel again.	Activate all internal and external commands.	When the “service” mode is switched off, the door status will be displayed.				

15. Trouble shooting

15.1. Type and position of the fuses at the control box

15.1.1. 230VAC Single phase

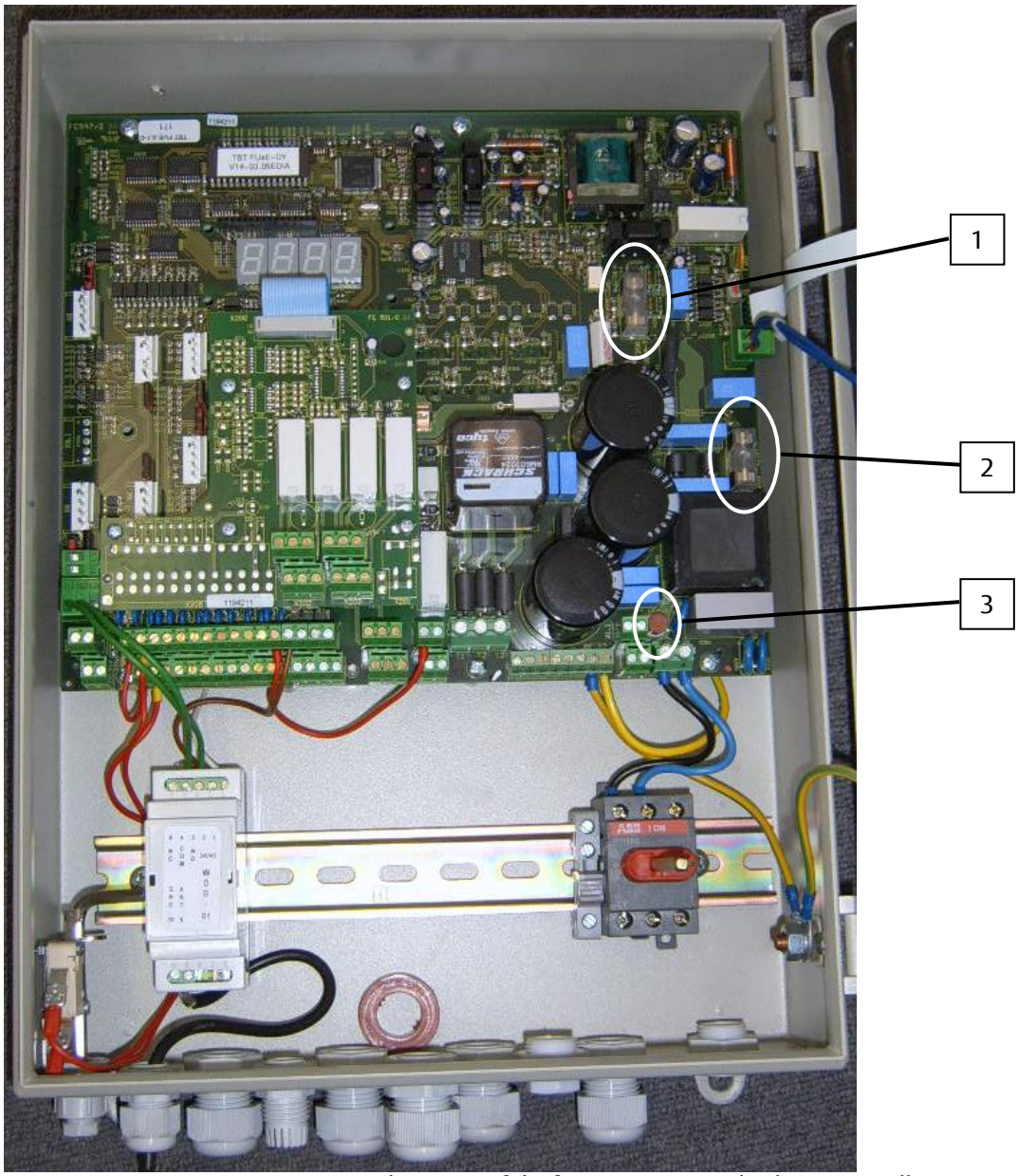
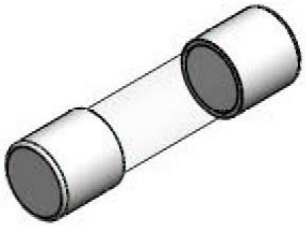
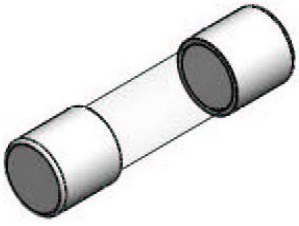
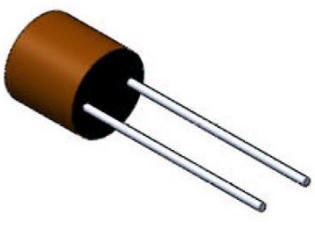


Figure 15-1: Type and position of the fuses 230VAC Single phase Controller.

		
1: T250mA L250V	2: T500mA L250V	T3, 15A 250V

15.1.2. 400VAC Three phase

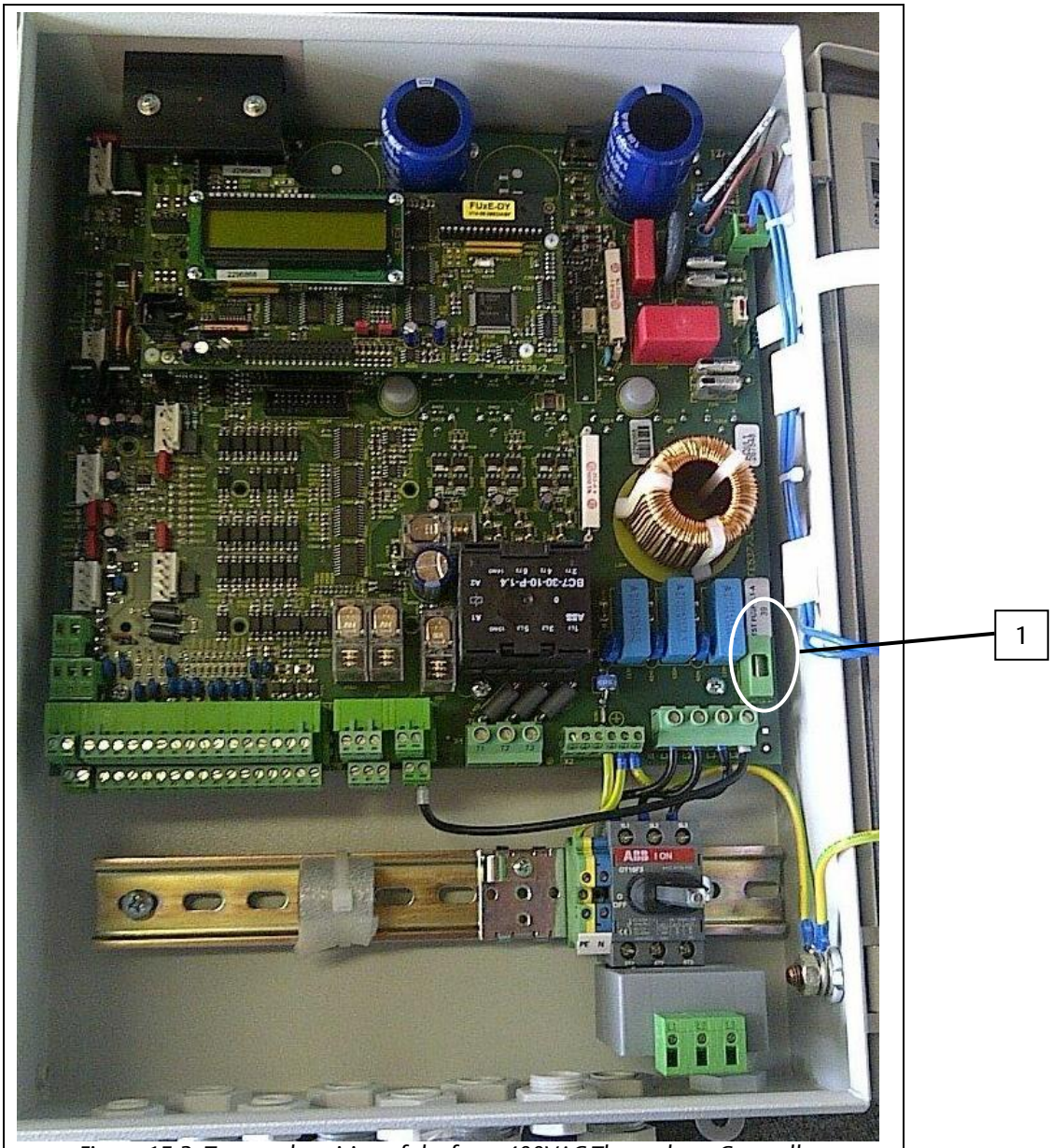
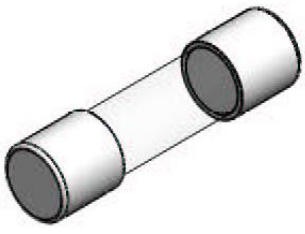


Figure 15-2: Type and position of the fuses 400VAC Three phase Controller.



1: T2, 5A L250V

15.2. Functions

P.	(Unit) range	Door functions	Default
000	(Cycles)	Door cycle counter display___ Display: 1234567 → 1234. -press .567 Display: 67 → 67	
010	(s) 0....200	Open hold time (End position Upper – Eo) 0 = automatic closing deactivated.	4
011	(s) 0....200	Open hold time (intermediate stop – E1) 0: Automatic closing deactivated	10
020	(ms) 0....1000	Pre-warning time before open The opening move is delayed following receipt of an OPEN command by the time specified in this parameter. Be careful to the unit. If you want to set a pre-warning to 4 seconds, then modify parameter P.020 to 400	0
025	(s) 0....20	Pre-warning time before close. The closing move is delayed following receipt of a CLOSE command or after expiration of the auto close time (forced close) by the time specified in this parameter	0

P	(Unit) range	Nominal motor data	Default
100	(Hz) 30...200	Motor rated frequency (see nameplate, note: Y/Δ)	-1
101	(A) 0...9.9	Motor rated current (see nameplate, note: Y/Δ)	-1
102	(%) 40...100	Power factor cosφ (see nameplate: cos φ: 0.63 →63)	-1
103	(V) 100...500	Motor rated voltage (see nameplate, note: Y/Δ). The motor characteristics curve is automatically calculated based on the rated frequency and nominal voltage.	-1
130	0...1	Motor rotational field 0: Right rotating 1: Left rotating	1

P	(Unit) range	Teaching the end positions with electronic limit switches	Default
210	0....5	Selecting the position calibrated by a deadman move ("teach in"): 0: no → None/Cancel 1: Eu → Lower and Upper limit switch (intermediate stop: see P244) 2: Eo → Upper limit switch (intermediate stop: see P244) 3: uo → Upper and Lower limit switch 4: E1 → Intermediate stop limit switch (P244 is ignored) 5: al → (all) Lower, Upper and Intermediate Stop limit switch (P244)	O

P	(Unit) range	Correcting end positions with electronic limit switches	Default
221	(Ink) ±125	Correction value for Lower end positition (set to 0 after new calibration).	0
231	(Ink) ±60	Correction value for Upper end position (set to 0 after new calibration).	0

P	(Unit) range	Diagnostics	Default
910	0…15	Selecting display mode (request by STOP button or during motor movement) 0: Controller sequence (automatic mode) 1: {Hz} Current rotary field frequency 2: {A} Current motor current (> 1A) 3: {V} Current motor current 4: {A} Current link current (effective current) 5: {V} Link voltage 6: {°C} End temperature in °C 7: {°F} End temperature in °F 8: Last measured run time (1/10 t/m 99.9s, 1/1 from 100s) For electronic limit switches only: 9: [Inc] current position progress 10: [Inc] current reference position 11: [dig] current channel 1 value of absolute encoder 12: [dig] current channel 2 value of absolute encoder 13: [dig] current reference voltage (2.5V) 14: Temperature in housing in (°C) 15: Temperature in housing in (°F)	0
920	Eb1 Eb2 Eb3 Eb4 Ebcl Eb	Display error memory/faults ⇒ Open by pressing Stop key again ⇒ Change by pressing Open/Close key ⇒ Quit by pressing Stop key. ⇒ Exit using cancel “EB-“ • Eb 1 → error messages 1 (most recent) or Er-- • Eb 2 → error messages 2 Er-- • Eb 3 → error messages 3 Er-- • Eb 4 → error messages 4 Er-- • Ebcl→ clear entire error memory • Eb - → Cancel (Display noEr: no error).	Eb 1
925		Software version display	
930	{s} 0….120.0	Motor run-time during last door move	
940	{V}	Line supply voltage display	
980	0….2	Service mode 0: Automatic (open and close in hold function) 1: Deadman close (manual mode close / automatic mode open) 2: Deadman (manual mode for open and close) 3: Emergency (deadman open and close; all errors and securities are ignored)	0

15.3. Overview of messages

Incorrect end positions		
F.000	Door position too far up	Too small a parameter value for upper emergency limit switch. Upper limit switch range (limit switch band) too small. Mechanical brake defective or improperly set.
F.005	Door position too far down	Too small a parameter value for lower emergency limit switch. Lower limit switch range (limit switch band) too small. Mechanical brake defective or improperly set.

Inaccuracies related to door movements		
F.020	Operating time exceeded (during opening, closing or dead man's use)	- Current motor operating time has exceeded the set maximum operating time; the door may not run smoothly or may be blocked. - One of the end stops has not been activated when mechanical end stops are used.
F.030	Dragging errors (position change of the door is less than expected)	- The door or motor is blocked - Too little output for the required torque - Too little speed - Attachment to the shaft of the absolute value detector or the increase detector is not secure - Incorrect position system in use (P.205).
F.031	The registered turning direction deviates from the expected turning direction	- When increase detector is used: Channel A and B have been exchanged - Motor rotation direction was changed at calibration - Too much "sagging" at the start, brakes start too early or the torque is insufficient; adjust the voltage if required
F.043	Fault of the front end stop of the light bar	- The front end stop for the light bar also stays in the centre end position and in the top end position, respectively.

Error message from the increase detector		
F.050	Position of the reference stop is not within the acceptable range during cyclic synchronisation.	- Reference stop is activated each time (defective). - Reference stop is activated too far from the selected reference. - Reference stop is activated by the belt of the end stop. - P270 and P280 are both on the reference stop.
F.051	Position of the reference stop is not within the acceptable range.	- Reference stop is located by the belt of the end stop. - Referenace stop is located outside 15% EO. - Reference stop is defective.
F.052	Reference stop is not recognized.	- During the synchronisation after activation, the reference stop is not recognized within 20% EO. - The reference stop is not recognized in the end position.

Parameter not set		
F.090	Parameters have not been allocated to the control.	The basic parameters (P.205, P.100 to P.103) of the TST FUE control have not yet been set.

Safety chain faults		
F.201	Internal E stop "push button" tripped or Watchdog (computer monitor).	E-stop chain was interrupted starting at input "internal E-stop" without parameterizing mode having been selected Internal parameter or EEPROM checks defective, pressing the STOP key provides additional information about the cause
F.211	External E-stop 1 tripped.	The E-stop chain was interrupted starting at input 1. (terminals 1 and 2)
F.212	External E-stop 2 tripped.	The E-stop chain was interrupted starting at input 2. (terminals 3 and 4)

Faults in the safety chain		
F.360	Short circuit detected on edge input	Short circuit detected on edges with normally closed contact
F.361	Number of edge trips for closing has reached set limit	Parameterized, maximum number of bottom edge trips during a door cycle was exceeded. Remove the object. Check the function of WDD
F.362	Redundancy error with short circuit	One of the processing channels for short circuit detection does not react identically with the second channel. → Controller board defective
F.363	Interruption on edge input	Connection cable defective or not connected Termination resistor incorrect or missing Jumper J600 incorrectly set
F.364	Safety edge testing failed	Safety edge was not activated as expected when requesting a test. The time between request for testing and actual testing not in agreement
F.365	Redundancy error with interruption	One of the processing channels for interruption detection does not react identically with the second channel. → Controller board defective
F.369	Internal safety edge incorrectly parameterized	An internal safety edge is connected but deactivated
F.374	Safety bar testing failed	Pre-limit switch for safety edge incorrectly set or defective Processing module defective Safety edge defective
F.385	Fault in pre-limit switch for safety edge	Pre-limit switch for turning off the safety edge or reversing after safety edge tripping remains tripped even in the upper end position.
F.400	Controller hardware reset detected	• Excessive noise on supply voltage • Internal watchdog tripped • RAM error
F.410	Over-current (motor current or intermediate circuit)	• Wrong motor data set (P100 – P103) • Non-adjusted voltage increase / boost set (P140 or P145) • Motor not properly dimensioned for door • Door sticks

Faults in the safety chain		
F.420	Overvoltage in intermediate circuit Limit 1	• Brake chopper interference / defective / missing • Feed voltage much to high • Motor feeds back too much energy in generator mode, door motion energy cannot be sufficiently brought down
F.430	Temperature cooler outside of working range Limit 1	• Excessive load on final stages or brake chopper • Ambient temperature too low for controller operation • Clock frequency of final stage too high (Parameter **P.160**)
F.435	Fault: Temperature in housing rises over 75 °C	• Excessive load on frequency converter / circuit • Controller cabinet insufficiently cooled
F.440	Overcurrent in intermediate circuit Limit 1	• Boost not adjusted • Motor incorrectly dimensioned for door • Door sticks
F.510	Motor / intermediate circuit overcurrent Limit 2	• Wrong motor data set (P100 – P103) • Non-adjusted voltage increase / boost set (P140 or P145) • Motor not properly dimensioned for door • Door sticks
F.511	Pb with injection of current for braking	• Check parameter P155 til P159, • Try to reduce P155 P156
F.515	Motor protection function detected overcurrent	• Incorrect motor curve (motor rated current) set (P101) • Too much boost (P140 or P145) • Motor incorrectly dimensioned
F.519	IGBT driver chip detected overcurrent	• Short circuit or ground fault on motor terminals • Motor rated current setting extremely wrong (P100) • Extremely too much boost (P140 or P145) • Motor incorrectly dimensioned • Motor winding defective • Momentary interruption of the E-Stop circuit.
F.520	Overvoltage in intermediate circuit Limit 2	• Brake chopper interference / defective / missing • Feed voltage much to high • Motor feeds back too much energy in generator mode, door motion energy cannot be sufficiently brought down.
F.521	Overvoltage in intermediate circuit	• Input voltage supply too low, usually at load • Load too great / final stage or brake chopper fault
F.524	Ext. 24 V supply missing or too low	• Overload but no short circuit • When 24V is shorted the controller voltage does not ramp up and glow lamp V306 comes on.
F.530	Temperature cooler outside of working range Limit 1	• Excessive load on final stages or brake chopper • Ambient temperature too low for controller operation • Clock frequency of final stage too high (Parameter **P.160**)
F.535	Fault: Temperature in housing rises over critical 80 °C	• .Internal temperature too high
F.540	Overcurrent in intermediate circuit Limit 2	• Boost not adjusted • Motor incorrectly dimensioned for door • Door sticks

Faults in the safety chain		
F.700	Position sensing defective	**nic limit switches**: • After invoking activation of the factory parameters (Parameter **P.990**) the corresponding positioning system was not parameterized. • Calibration not completed or is incorrect and must be repeated. • When activating the intermediate stop the intermediate stop is implausible. • Synchronization not finished or reference switch defective.
F.720	Synchronization error in position sensing with encoder	• Intermediate stop position is less than the minimum incremental value (25). • Synchronization was not finished. • The selected reference switch was not reached or is outside its tolerance • The incremental encoder is not counting or the door is blocked (also F.030, lag error) • Incremental inputs IN 9 and IN 10 are reversed (also F.031 rotation error)
F.750	Protocol Transmission error	• Defective hardware or electrically noisy environment
F.751	Synchronization FU <-> absolute encoder	• Defective hardware or electrically noisy environment • Absolute encoder processor electronics defective
F.752	Timeout with protocol transmission	• Interface cable defective / interrupted • Absolute encoder processor electronics defective • Defective hardware or electrically noisy environment
F.758	Protocol transmission DNC2	• Battery of encoder is empty • Cable DNC2 is damaged.
F.760	Position outside of window	• Position encoder drive defective • Absolute encoder processing electronics defective • Defective hardware or electrically noisy environment
F.761	Distance Channel 1 <-> Channel 2 outside allowed window	• Position encoder drive defective • Defective hardware or electrically noisy environment
F.762	Electronic limit switch positions incorrect	• Upper limit switch Eo or intermediate limit switch E1 has exceeded the valid limit range • Controller not yet initialized • Position values during calibration not correct or values are no longer plausible

15.3.1. F.9xx internal system-dependent errors

	Internal errors that can still be resolved by the user are involved with this type of errors. When such an error occurs, contact customer service immediately.

15.3.2. Information messages

General messages	
STOP	Stop / Reset status, wait for next incoming command.
_Eu_	End position Lower Eu
≡Eu≡	End position Lower locked → no raising possible (e.g., lock-door)
ZUF°	Closing active
¯Eo¯	End position Upper Eo.
≡Eo≡	End position Upper locked → no closing possible (e.g., safety edge).
°AUF	Opening active.
-E1-	End position middle E1 (intermediate stop position)
≡E1≡	End position middle locked → no closing possible (e.g., safety edge)
FAIL	Fault → only deadman travel is possible, possibly automatic opening
CALI / EICH	Calibration → setting the end positions in deaman travel (for absolute encoder) → start procedure using STOP key.
≡NA≡	E-Stop → No travel possible, hardware safety chain interrupted.
NOTF	E-travel → Deadman travel without taking into account safeties, etc.
'Hd'	Manual → Deadman mode.
ParA	Parameterizing
SYNC	Synchronization (incremental encoder/limit switch → pos. unknown).
'Au'	Automatic → indicates change from “Manual” to “Automatic” status
'Hc'	Semi-automatic → indicates change from “Manual” to “Semi-automatic”.
FUS	Initial display after power up. (Power Up and self-test)

Status messages during calibration	
E.i.E.u.	Calibration of the end position below the requested (in dead man's use).
E.i.E.o.	Calibration of the end position above the requested (in dead man's use).
E.i.E.1.	Calibration of the E1 intermediate position (in dead man's use).

Status messages during synchronisation	
S.y.E.u.	Synchronization of End position Lower requested (deadman or wait for start condition)
S.y.E.o.	Synchronisation of End position Upper requested (deadman or wait for start condition)
S.y.E.1.	Synchronisation of intermediate stop position E1 (in deadman)
S.y.op	Automatic open until mechanical stop, then automatic synchronization of End position Upper
S.y.Cl	Automatic close taking into account safeties until mechanical stop, then automatic synchronization of End position Lower
S.y.c≡	Automatic close is locked due to request A

Status messages during dead man's use	
Hd.cL	Dead man's use when closing (membrane button: CLOSE).
Hd.oP	Dead man's use when opening (membrane button: OPEN).
Hd.Eu	End position under range; no more dead man's use when closing possible.
Hd.Eo	End position above the range; no more dead man's use when opening possible.
Hd.Ao	Outside the allowed Eo position (dead man's use not possible when opening).

Information messages during automatic use	
I.080	Maintenance required soon / service counter nearly expired.
I.100	Speed when reaching upper end position too high.
I.150	Speed when reaching lower end position too high.
I.160	Continuous CLOSED still active
I.170	Forced opening being performed.
I.199	Door cycle not plausible (re-initialize → Parameters)
I.200	Reference position corrected or recognized, (after calibration).
I.201	Reference position re-initialized.
I.202	Reference position missing.
I.203	Reference position incorrect.
I.205	Synchronization.
I.210	Pre-limit switch Upper not plausible.
I.211	Pre-limit switch Lower not plausible.
I.310	Open-command on Door2 being issued (AIRLOCK)
I.500	Correction of upper limit switch running
I.501	Upper pre-limit switch corrected
I.502	Upper limit switch band corrected
I.505	Correction of lower limit switch running.
I.506	Lower pre-limit switch corrected
I.507	Lower limit switch band corrected
I.510	Limit switch correction finished
I.515	Controller is preparing automatic teach-in of the limit switches
I.555	Limit switches being corrected

Information messages during parameter allocation	
noEr	Memory error: an error will not be saved.
Er--	Memory error: could not find the corresponding message for the error.
Prog	Programming message when the original parameter or default setting, respectively, was being executed.

General inputs	
E.000	Open membrane button.
E.050	Stop membrane button.
E.090	Close membrane button.
E.101	Input 1 / terminal 6
E.102	Input 2 / terminal 8
E.103	Input 3 / terminal 10
E.104	Input 4 / terminal 12
E.105	Input 5 / terminal 14
E.106	Input 6 / terminal 16
E.107	Input 7 / terminal 18
E.108	Input 8 / terminal 20
E.109	Input 9 / terminal 22
E.110	Input 10 / terminal 24
E.132	Check whether PA32 has a value of 0.

Safety range emergency range	
E.201	Internal Emergency OFF button activated.
E.211	External Emergency stop 1 activated.
E.212	External Emergency stop 2 activated.

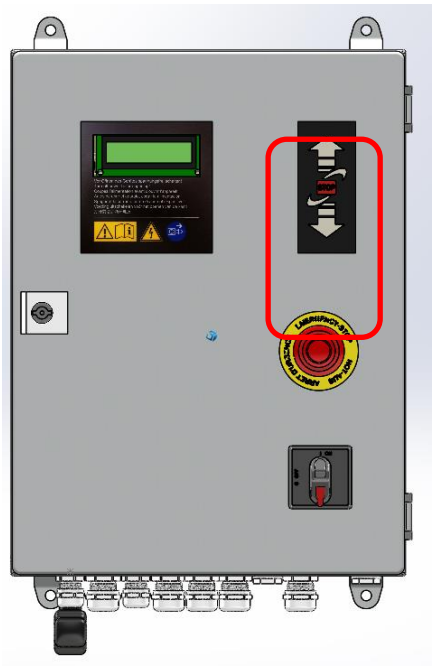
Tracer in general	
E.360	Activation of internal safety edge.
E.363	Fault in internal safety edge
E.370	Activating of external safety edge.
E.373	Fault in external safety edge.
E.379	External safety edge activated but not yet plugged in.

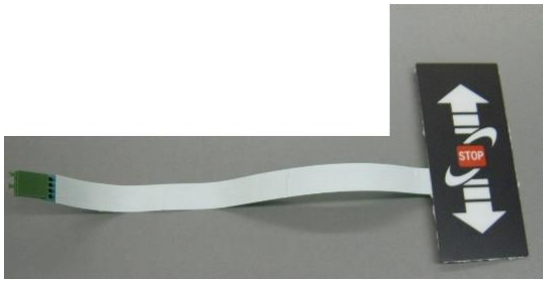
Induction loop processor plug_in module	
E.501	Detector channel 1
E.502	Detector channel 2

Internal Inputs	
E.900	Error signal of the drive module.

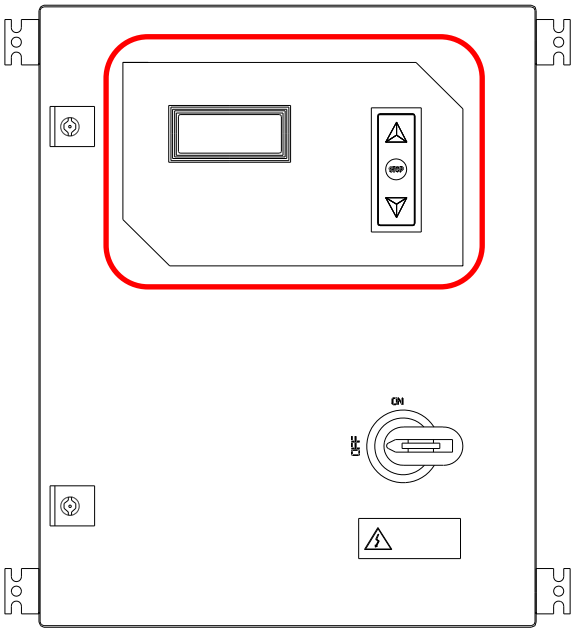
16. Part submitted to wear

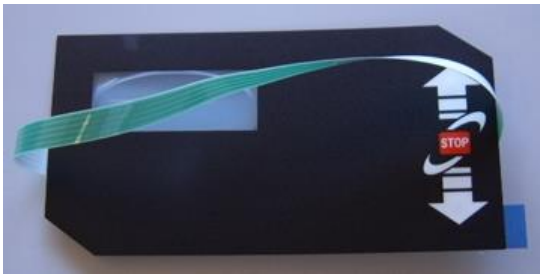
16.1. Steel controller





16.2. Stainless steel controller





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